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EIQP: Execution-time-certified and Infeasibility-detecting QP Solver

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***Abstract*—Solving real-time quadratic programming (QP) is a ubiquitous task in control engineering, such as in model predictive control and control barrier function-based QP. In such real-time scenarios, certifying that the employed QP algorithm can either return a solution within a predefined level of optimality or detect QP infeasibility before the predefined sampling time is a pressing requirement. This article considers convex QP (including linear programming) and adopts its homogeneous formulation to achieve infeasibility detection. Exploiting this homogeneous formulation, this article proposes a novel infeasible interior-point method (IPM) algorithm with the best theoretical $O(\sqrt{n})$ iteration complexity that feasible IPM algorithms enjoy. The iteration complexity is proved to be *exact* (rather than an upper bound), *simple to calculate*, and *data independent*, with the value $\left\lceil \frac{\log(\frac{n+1}{\epsilon})}{-\log(1-\frac{0.414213}{\sqrt{n+1}})} \right\rceil$ (where n and ϵ denote the number of constraints and the predefined optimality level, respectively), making it appealing to certify the execution time of online time-varying convex QPs. The proposed algorithm is simple to implement without requiring a line search procedure (uses the full Newton step), and its C-code implementation (offering MATLAB, Julia, and Python interfaces) and numerical examples are provided.**

***Index Terms*—Quadratic programming, execution time certificate, infeasibility detection, model predictive control, control barrier function.**

I. Introduction

Like solving a linear system, solving a quadratic programming (QP) problem has also been ubiquitous and extensively used including in finance, machine learning, operations, computer vision, energy, transportation, bioinformatics, signal processing, robotics, and control. For example, *i)* model predictive control (MPC), which formulates a receding horizon control problem as a QP that is solved at each sampling time, has been widely spread in the industry for controlling multivariable systems subject to constraints [1] and *ii)* CLF-CBF-QP [2], a control framework that combines stability (via control Lyapunov function) and safety (via control barrier functions) into a real-time QP, has been widely used in autonomous systems. Specifically regarding applications, *i)* the quadrotor maneuvering field has a well-known QP-based framework [3] for generating minimum snap trajectories with smooth transitions through waypoints while satisfying constraints on velocities, accelerations, and inputs; *ii)* the robotics field often uses QP as a unified framework for task-space control paradigms [4], [5]. All these QPs fall into the categories of *online*, as they are required to be executed in real time, in contrast to the *offline* QPs, which do not have this requirement.

Solving *real-time* QPs poses a pressing and open challenge for algorithms: *how to guarantee that the QP algorithm can return the optimal solution or detect infeasibility before the predefined sampling (or feedback) time*. This is called the *execution time certificate* problem. Without the *execution time certificate*, control signals (the backup control signal if detecting QP infeasibility) fail to return on time, which renders the system open-loop and potentially leads to safety concerns, especially in safety-critical process systems.

Certifying the execution time of QP algorithms can be reduced to certifying the number of iterations of QP algorithms if each iteration performs a fixed number of floating-point operations ([flops]). In fact, nearly all QP solvers have default settings that specify a maximum number of iterations, seemingly addressing the worst-case *execution time certificate* problem. However, this is a *cheating* approach for addressing *execution time certificate* problem, because it can not guarantee that all returned solutions have *the same level of optimality*.

An execution-time-certified QP algorithm should precisely be defined as one that *either returns an optimality-certified solution (within a predefined level of optimality) or detects QP infeasibility before the predefined sampling time*. Recently, the *execution time certificate* problem has attracted significant scholarly interest and remains a vibrant research area, especially within the context of MPC [6]–[21].

A. Related work

Current QP algorithms tailored for efficiently solving MPC include first-order, active set, and interior point methods [22]–[26]. The iteration complexity analysis of first-order methods is provided in [8]–[10] for linear MPC problems, but their iteration complexity result is dependent on the QP problem data, such as the Hessian matrix. Their *data-dependent* iteration complexity result cannot guarantee the time-invariant number of iterations for MPC-based QP with a time-varying Hessian matrix, such as Real-Time Iteration (RTI)-based nonlinear MPC [27].

Although active set methods often have good performance in small- to medium-scale problems in practice, in theory, they can have an exponential number of iterations in the worst case [28]. Refs. [11]–[15] develop a computationally complicated and expensive (thus offline) worst-case partial enumeration technique, to certify the worst-case number of iterations for active set methods. Their iteration complexity results are also *data-dependent* and cannot be used in QPs that have time-varying Hessian matrices. This also happens in work [16], which proposes an iteration-complexity-certified active-set algorithm for MPC problems. Although its iteration complexity is only dimension-dependent, it relies on the data-dependent modified N-step vector, whose search process is a linear programming problem, and thus limits its application to real-time QPs that have a time-varying Hessian matrix. Their

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subsequent work [17] eliminates the need for the N-step vector by introducing perturbations to the constraints and solving a sequence of LCP problems, providing an explicit upper bound on computational complexity; however, the iteration complexity still depends on the problem data.

Interior point methods (IPMs) can be categorized as being *practical* (such as Mehrotra predictor-corrector IPMs [29]) or *theoretical*. Practical Mehrotra predictor-corrector IPMs behave super fast (with usually < 50 iterations in practice) and became the foundation for most interior point software, such as [30], but are heuristic and may diverge for some examples [31, see p. 411], [32], without certified global convergence proofs. *Theoretical* IPMs can be classified as being infeasible or feasible, which generally have certified $O(n)$ and $O(\sqrt{n})$ iteration complexity [33], [34], respectively. However, the faster iteration complexity of feasible IPMs comes at the cost of requiring a strictly feasible initial point lying in the neighborhood of the central path. Finding this strictly feasible initial point for a general QP usually requires solving an LP, known as the *phase I* stage [35, Sec. 11.4]. To avoid this limitation, our previous work [18] for the first time proposed a *cost-free* initialization strategy for feasible IPMs by exploiting the structure of the considered box-constrained QP (arising from input-constrained MPC). One interesting feature of our algorithm [18] is that its iteration complexity is the *exact* value: $\left\lceil \frac{\log(\frac{2n}{\epsilon})}{-2\log(\frac{\sqrt{2n}}{\sqrt{2n}+\sqrt{2}-1})} \right\rceil + 1$, rather than an upper bound as provided in previous works. This *exact* iteration complexity is *data independent* (only *dimension dependent*), so our algorithm can certify the iteration of QP with time-varying data [20]. Although our algorithm [18] is only valid for box-constrained QP, Ref. [21] then extends it to general strictly convex QPs via an ℓ_1-penalty soft constraint which is transformed to box-constrained QP via duality theory. This soft constraint approach can offer a viable solution when the strictly convex QP is infeasible, such as in cases where an MPC encounters unknown disturbances or modeling errors that render the MPC-based QP problem infeasible.

However, some scenarios require real-time QP solvers to have the ability to detect infeasibility. For example, the soft-constrained solution is always approximate and hard to guarantee the stability of MPC. In this case, the backup control law can be active when detecting the potential infeasibility of real-time QP. Naturally, detecting infeasibility should also be completed within the certified execution time; all the aforementioned iteration complexity works [8]–[20] do not have infeasibility-detection capability and either assume the QP is feasible or restrict their consideration to box-constrained QPs.

Regarding infeasibility detection in QP, *i)* the first-order method, specifically the alternating direction method of multipliers, has been reported to possess the infeasibility-detection capability in [36] and its corresponding solver OSQP [22], but lacks the iteration complexity analysis; *ii)* the active set method, combined with the nonnegative least-squares (NNLS) formulation, has infeasibility-detection capability in [37] and can be combined with the offline (expensive) iteration complexity certification procedure [38] to determine the worst-case iterative behavior, but this NNLS formulation is only limited to strictly convex QPs; *iii)* IPM, combined with the homogeneous slacked QP formulation [39], possesses the infeasibility-detection capability but is limited to strictly convex QPs and lacks a certified iteration complexity for the employed IPM algorithm. IPMs based on the homogeneous self-dual (HSD) formulation [40], [41] are well known to be capable of detecting the infeasibility of LPs. Furthermore, Ref. [42] extended the HSD formulation to the linear complementarity problem (LCP) (more general than LP), but their proposed IPM has *data-dependent* iteration complexity $O(\sqrt{n}L)$ (where L is the binary data length of LCP problem data). Ref. [43] proposed a homogeneous formulation for nonlinear monotone complementarity problems to support infeasibility detection, and provided iteration complexity analysis for the employed infeasible IPM, but its iteration complexity result is too conservative and *data dependent*, specifically depending on the scaled Lipschitz constants of the nonlinear monotone mapping. Ref. [44] adopts a homogeneous formulation for convex optimization problems with conic constraints, but lacks a theoretical iteration complexity analysis as it focuses on practical computation efficiency.

B. Contributions

This article extends the homogeneous formulation from [43] to detect the infeasibility of convex QPs and proposes a novel infeasible IPM algorithm to solve convex QPs. Our novel contributions are the following:

1) proves that this homogeneous formulation enables our proposed infeasible IPM algorithm to achieve the best theoretical $O(\sqrt{n})$ iteration complexity, which feasible IPM algorithms enjoy, so there is no need to find a strictly feasible initial point.
2) proves that the iteration complexity of our algorithm is *exact* (rather than the upper bound), *simple to calculate*, and *data independent*, with the value
$$\left\lceil \frac{\log(\frac{n+1}{\epsilon})}{-\log(1-\frac{0.414213}{\sqrt{n+1}})} \right\rceil$$
(where n and ϵ denote the number of constraints and the predefined optimality level, respectively), making it appealing to certify the execution time of online time-varying convex QPs (including LPs).
3) shows that our algorithm is *simple* to implement as it does not require a line-search procedure. Its C-code implementation is just one single C file, making it easy to integrate with other software and appealing when deployed in embedded production systems. Its MATLAB, Julia, and Python interfaces and numerical examples are publicly available at https://github.com/liangwu2019/EIQP. Moreover, our algorithm is *parameter free* as there are no algorithm parameters, avoiding the tuning work.

To the best of the authors' knowledge, this article for the first time provides a convex QP solver with *exact* and *data-independent* **E***xecution time certificate* and **I**nfeasibility detection, thus referred to as EIQP.

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C. Notations

$\lceil x \rceil$ maps x to the least integer greater than or equal to x. $\mathbb{R}^n$ denotes the space of n-dimensional real vectors, $\mathbb{R}^n_+$ and $\mathbb{R}^n_{++}$ are the set of all non-negative and positive vectors of $\mathbb{R}^n$, respectively. $Q \succ 0$ ($Q \succeq 0$) denotes positive definiteness (semi-definiteness) of a square matrix Q, $Q^\top$ (or $z^\top$) denotes the transpose of matrix Q (or vector z). $\min(z, y)$ denotes the minimum value between scalar z and y. Given two arbitrary vectors $z, y \in \mathbb{R}^n_{++}$, their Hadamard product is $z \odot y = (z_1 y_1, z_2 y_2, \cdots, z_n y_n)^\top$, $\frac{z}{y} = z \odot \left(y^{-1}\right) = \left(\frac{z_1}{y_1}, \frac{z_2}{y_2}, \cdots, \frac{z_n}{y_n}\right)^\top$. For $z, y \in \mathbb{R}^n$, let $\mathrm{col}(z, y) = (z^\top, y^\top)^\top$ and $\max(z, y) = (\max(z_1, y_1), \max(z_2, y_2), \cdots, \max(z_n, y_n))^\top$. For a vector $z \in \mathbb{R}^n$, its Euclidean norm is $\|z\| = \sqrt{z_1^2 + z_2^2 + \cdots + z_n^2}$, $\|z\|_1 = \sum_{i=1}^n |z_i|$, $\|z\|_\infty = \max_i |z_i|$, $\mathrm{diag}(z) : \mathbb{R}^n \to \mathbb{R}^{n \times n}$ maps a vector z to its corresponding diagonal matrix, $\sqrt{z} = \left(\sqrt{z_1}, \sqrt{z_2}, \cdots, \sqrt{z_n}\right)^\top$ and $\frac{1}{z} = (\frac{1}{z_1}, \frac{1}{z_2}, \cdots, \frac{1}{z_n})^\top$. The vector of all ones is denoted by $e = (1, \cdots, 1)^\top$.

II. Problem formulations

This article considers the convex QP,

$$\begin{aligned} \min_z \ & \frac{1}{2} z^\top Q(t) z + c(t)^\top z \\ \text{s.t. } & A(t) z \geq b(t), \\ & z \geq 0 \end{aligned} \tag{1}$$

where the data $Q(t) = Q(t)^\top \in \mathbb{R}^{n_z \times n_z} \succeq 0$, $c(t) \in \mathbb{R}^{n_z}$, $A(t) \in \mathbb{R}^{n_b \times n_z}$, $b(t) \in \mathbb{R}^{n_b}$, are all often time-varying in real-time applications. Problem (1) is common such as in real-time MPC with state and input constraints and CLF-CBF-QP with input bounds, see subsections VI-A and VI-B. For example, a typical MPC or CLF-CBF-QP problem, $\min_z \frac{1}{2} z^\top Q(t) z + c(t)^\top z$, s.t. $lb(t) \leq z \leq ub(t), A(t) z \leq b(t)$, (even its soft-constrained formulation), can be easily formulated as Problem (1) by letting $z \leftarrow z - lb(t)$, $c(t) \leftarrow c(t) + Q(t) lb(t)$, $A(t) \leftarrow [-I; -A(t)]$, and $b(t) \leftarrow [lb(t) - ub(t); A(t) lb(t) - b(t)]$. A standard QP formulation can be reformulated as (1).

Lemma 1. *Consider the standard QP formulation,*

$$\begin{aligned} \min_z \ & f(z) \triangleq \frac{1}{2} z^\top Q(t) z + c(t)^\top z \\ \text{s.t. } & z \in S \triangleq \{z \in \mathbb{R}^{n_z} : A(t) z \geq b(t)\} \end{aligned} \tag{2}$$

and its corresponding QP:

$$\begin{aligned} \min_{\bar{z}} \ & f'(\bar{z}) \triangleq \frac{1}{2} \bar{z}^\top \bar{Q}(t) \bar{z} + \bar{c}(t)^\top \bar{z} \\ \text{s.t. } & \bar{z} \in S' \triangleq \{\bar{z} \in \mathbb{R}^{2n_z} : \bar{A}(t) \bar{z} \geq b(t), \ \bar{z} \geq 0\} \end{aligned} \tag{3}$$

where $\bar{z} \triangleq \mathrm{col}(z_+, z_-)$,

$$\bar{Q}(t) = \begin{bmatrix} Q(t) & -Q(t) \\ -Q(t) & Q(t) \end{bmatrix} \succeq 0 \ (\textit{even if } Q(t) \succ 0),$$

$\bar{c}(t) = \mathrm{col}(c(t), -c(t))$, *and* $\bar{A}(t) = [A(t), -A(t)]$.

Let $\bar{z}^* = \mathrm{col}(z_+^*, \ z_-^*)$ *be a minimizer of QP* (3)*, then* $z^* = z_+^* - z_-^*$ *is a minimizer of QP* (2).

Proof. First, note that $z^* \in S$ because $\bar{z}^* = \mathrm{col}(z_+^*, \ z_-^*) \in S'$ implies $A(t) z^* \geq b(t)$. Also, by the definition of f'

$$f'(\bar{z}^*) = \frac{1}{2}(z_+^* - z_-^*)^\top Q(t)(z_+^* - z_-^*) + c(t)^\top (z_+^* - z_-^*) = f(z^*).$$

We now prove $z^* = z_+^* - z_-^*$ is a minimizer of QP (2) by contradiction. Suppose that z^* is not a minimizer of f on S. Then, there exists $z' \in S$ such that

$$f(z') < f(z^*).$$

We then construct the canonical nonnegative split of z':

$$z'_+ \triangleq \max(z', 0), \ z'_- \triangleq \max(-z', 0)$$

and $\bar{z}' = \mathrm{col}(z'_+, z'_-)$.

By construction $z'_+ \geq 0$, $z'_- \geq 0$, and $z'_+ - z'_- = z'$. Since $z' \in S$ we have $\bar{A}(t) \bar{z}' = A(t)(z'_+ - z'_-) = A z' \geq b$, so $\bar{z}' \in S'$. Moreover,

$$\begin{aligned} f'(\bar{z}') &= \frac{1}{2}(z'_+ - z'_-)^\top Q(t)(z'_+ - z'_-) + c(t)^\top (z'_+ - z'_-) \\ &= \frac{1}{2} z'^\top Q(t) z' + c(t)^\top z' \\ &= f(z') < f(z^*) = f'(\bar{z}^*), \end{aligned}$$

which contradicts the assumption that $\bar{z}^*$ is a minimizer of f' on S'. This completes the proof. □

Remark 1. *When* $Q = 0$*, Problem* (2) *becomes an LP. It is emphasized that the proposed Algorithm 2 (see Section IV-C), with execution time certificate and infeasibility detection, is also valid for solving LP.*

A. KKT condition and LCP formulation

The Karush–Kuhn–Tucker (KKT) condition [35, Ch. 5] for (1) is

$$\begin{aligned} & v = Q(t) z - A(t)^\top y + c(t) \\ & w = A(t) z - b(t) \\ & v \odot z = 0, v \geq 0, z \geq 0 \\ & w \odot y = 0, w \geq 0, y \geq 0 \end{aligned} \tag{4}$$

where $y \in \mathbb{R}^{n_b}$ denotes the Lagrangian variable for inequality constraints $A(t) z \geq b(t)$, $v \in \mathbb{R}^{n_z}$ denotes the Lagrangian variable for $z \geq 0$, and $w \triangleq A(t) z - b(t) \in \mathbb{R}^{n_b}$ is the slack variable. Eq. (4) is an LCP; for simplicity, we adopt the notations,

$$x \triangleq \mathrm{col}(z, y) \in \mathbb{R}^n, \ s \triangleq \mathrm{col}(v, w) \in \mathbb{R}^n,$$

where the dimension $n = n_z + n_b$. Then Eq. (4) is reformulated as

$$\text{LCP} \ \begin{cases} s = f(x) \triangleq M x + q \\ x \odot s = 0 \\ x \geq 0, s \geq 0 \end{cases} \tag{5}$$

where

$$M \triangleq \begin{bmatrix} Q(t) & -A(t)^\top \\ A(t) & 0 \end{bmatrix}, \quad q \triangleq \begin{bmatrix} c(t) \\ -b(t) \end{bmatrix}.$$

Note that $M \neq M^\top$. $M \succeq 0$ as $M + M^\top \succeq 0$ (easy to prove as $Q(t) \succeq 0$).

The QP (1) may be infeasible, such that the set $\{z \in \mathbb{R}^{n_z} : A(t) z \geq b(t), z \geq 0\}$ is empty and there is no solution for the LCP (5). To achieve infeasibility detection, this paper adopts a homogeneous LCP formulation from [43] and LCP (5) is a linear case of the nonlinear monotone complementarity problem [43], by the following Lemma.

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Lemma 2. *Let $g(x) : \mathbb{R}^n \to \mathbb{R}^n$ be differentiable on $\mathbb{R}^n$. If its symmetric part of the Jacobian,*

$$\frac{J_g(x) + J_g(x)^\top}{2} \succeq 0,$$

then $g(x)$ is a continuous monotone mapping in $\mathbb{R}^n$:

$$\forall x_1, x_2 \in \mathbb{R}^n,\ (x_1 - x_2)^\top (g(x_1) - g(x_2)) \geq 0.$$

Proof. Given two arbitrary $x_1, x_2 \in \mathbb{R}^n$. Consider the line segment $w(\theta) \triangleq x_2 + \theta(x_1 - x_2)$ where $\theta \in [0, 1]$. By the mean value theorem for vector-valued functions,

$$g(x_1) - g(x_2) = \int_0^1 J_g(w(\theta))(x_1 - x_2) d\theta$$

and take the inner product with $x_1 - x_2$:

$$\begin{aligned}&(x_1 - x_2)^\top (g(x_1) - g(x_2))\\ &= \int_0^1 (x_1 - x_2)^\top J_g(w(\theta))(x_1 - x_2) d\theta,\end{aligned}$$

And for any matrix A and vector v,

$$v^\top A v = v^\top \frac{A + A^\top}{2} v,$$

so each integrand equals

$$(x_1 - x_2)^\top \left(\frac{J_g(w(\theta)) + J_g(w(\theta))^\top}{2} \right) (x_1 - x_2),$$

which is nonnegative for all $\theta \in [0, 1]$ from the assumption $(J_g(x) + J_g(x)^\top)/2 \succeq 0$, Therefore, the integral is nonnegative, proving monotonicity $(x_1 - x_2)^\top (g(x_1) - g(x_2)) \geq 0$. Finally, differentiability of $g(x)$ implies continuity of $g(x)$, so $g(x)$ is continuous, which completes the proof. □

By Lemma 2, and since the Jacobian of $f(x)$, denoted M, is positive semi-definite, we obtain

Corollary 1. *$f(x)$ in LCP (5) is a continuous monotone mapping in $\mathbb{R}^n_+$.*

B. Preliminaries: Homogeneous LCP formulation

The main results of this subsection are drawn from [43], which form the basis for the forthcoming algorithm descriptions.

By introducing two additional scalars $\tau \in \mathbb{R}_+$ and $\kappa \in \mathbb{R}_+$, a homogeneous LCP (HLCP) is formulated as

$$\text{HLCP} \begin{cases} \begin{bmatrix} s \\ \kappa \end{bmatrix} = \psi(x, \tau) \\ \begin{bmatrix} x \\ \tau \end{bmatrix} \odot \begin{bmatrix} s \\ \kappa \end{bmatrix} = 0 \\ (x, \tau, s, \kappa) \geq 0 \end{cases} \tag{6}$$

where

$$\psi(x, \tau) \triangleq \begin{bmatrix} Mx + q\tau \\ -x^\top M x/\tau - x^\top q \end{bmatrix} : \mathbb{R}^{n+1}_{++} \to \mathbb{R}^{n+1}. \tag{7}$$

Note that the HLCP (6) inherits the monotone feature by the following Lemma.

Lemma 3. *The Jacobian of $\psi(x, \tau)$ is*

$$J_\psi(x, \tau) = \begin{bmatrix} M & q \\ -x^\top (M + M^\top)/\tau - q^\top & x^\top M x/\tau^2 \end{bmatrix} \tag{8}$$

and $J_\psi(x, \tau)$ is positive semi-definite in $\mathbb{R}^{n+1}_{++}$.

Thus, by Lemma 2, $\psi(x, \tau)$ is a continuous monotone mapping in $\mathbb{R}^{n+1}_{++}$.

Proof. By the definition of $\psi(x, \tau)$ (7), its Jacobian matrix $J_\psi(x, \tau)$ indeed follows (8).

Given an arbitrary $\mathrm{col}(d_x, d_\tau) \in \mathbb{R}^{n+1}$ (where $d_x \in \mathbb{R}^n$, $d_\tau \in \mathbb{R}$) and M is positive semi-definite,

$$\begin{aligned}&\left[d_x^\top, d_\tau\right] J_\psi(x, \tau) \begin{bmatrix} d_x \\ d_\tau \end{bmatrix}\\ &= d_x^\top M d_x + (d_x^\top q) d_\tau - \frac{d_\tau}{\tau} x^\top (M + M^\top) d_x - d_\tau q^\top d_x\\ &\quad + \frac{d_\tau}{\tau} x^\top M x \frac{d_\tau}{\tau}\\ &= \frac{1}{2} d_x^\top (M + M^\top) d_x - \frac{d_\tau}{\tau} x^\top (M + M^\top) d_x\\ &\quad + \frac{1}{2} \frac{d_\tau}{\tau} x^\top (M + M^\top) x \frac{d_\tau}{\tau}\\ &= \frac{1}{2} \left(d_x - \frac{d_\tau}{\tau} x \right)^\top (M + M^\top) \left(d_x - \frac{d_\tau}{\tau} x \right) \geq 0,\end{aligned} \tag{9}$$

and by Lemma 2, this completes the proof. □

Definition 1. *HLCP (6) is said to be asymptotically feasible if and only if there is positive and bounded iterates $(x^k, \tau^k, s^k, \kappa^k) > 0, k = 1, 2, \cdots$ such that $\lim_{k\to\infty} \mathrm{col}(s^k, \kappa^k) - \psi(x^k, \tau^k) \to 0$, where any limit point of the sequence $(x^*, \tau^*, s^*, \kappa^*)$ is called an asymptotically feasible point for HLCP (6). An asymptotically feasible solution $(x^*, \tau^*, s^*, \kappa^*)$ of HLCP (6) such that $(x^*)^\top s^* + \tau^* \kappa^* = 0$ is said to be an "optimal" or "complementary" solution of HLCP (6).*

Lemma 4. *HLCP (6) is always asymptotically feasible. Every asymptotically feasible solution of HLCP (6) is an "optimal" or "complementary" solution.*

Proof. Take $x^k = (\frac{1}{2})^k e, \tau^k = (\frac{1}{2})^k, s^k = (\frac{1}{2})^k e$, and $\kappa^k = (\frac{1}{2})^k$. Then, as $k \to \infty$,

$$\begin{bmatrix} s^k \\ \kappa^k \end{bmatrix} - \psi(x^k, \tau^k) = \left(\frac{1}{2}\right)^k \begin{bmatrix} e - Me - q \\ 1 + e^\top M e + e^\top q \end{bmatrix} \to 0,$$

which completes the first part of the proof.

Moreover, let $(x^*, \tau^*, s^*, \kappa^*) \geq 0$ be any (asymptotically) feasible solution for HLCP (6), then by Eqn. (7) we have

$$(x^*)^\top s^* + \tau^* \kappa^* = \mathrm{col}(x^*, \tau^*)^\top \psi(x^*, \tau^*) = 0,$$

which completes the second part of the proof. □

Definition 2. *An "optimal" or "complementary" solution (x^*, s^*) for LCP (5) (or $(x^*, \tau^*, s^*, \kappa^*)$) for HLCP (6)) is said to be a maximal complementary solution if the number of positive components in (x^*, s^*) (or in $(x^*, \tau^*, s^*, \kappa^*)$) is maximal.*

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Lemma 5. *(see [45, Theorem 2.3]): The indices for those positive components are invariant among all maximal complementary solutions for LCP* (5) *(or HLCP* (6)*).*

Thus, finding a maximal complementary solution for HLCP (6) is equivalent to finding a complementary solution for LCP (5), and the relationship between the solutions of HLCP (6) and LCP (5) can be described by the following Lemma.

Lemma 6. *(see [43, Thm. 1]): Let* $(x^*, \tau^*, s^*, \kappa^*)$ *be a maximal complementary for HLCP* (6)*. Then*

i) LCP (5) *has a solution if and only if* $\tau^* > 0$*. In this case,* $\mathrm{col}(x^*/\tau^*, s^*/\tau^*)$ *is a complementary solution for LCP* (5)*;*

ii) LCP (5) *is infeasible if and only if* $\kappa^* > 0$*. In this case,* $\mathrm{col}(x^*/\kappa^*, s^*/\kappa^*)$ *is a certificate to prove infeasibility.*

Thus, finding the solution or detecting the infeasibility of LCP (5) is equivalent to finding a maximal complementary solution of HLCP (6).

C. Central path of HLCP

Denote $\bar{r} \triangleq \mathrm{col}(r, \tilde{r}) \in \mathbb{R}^{n+1}$ (where $r \in \mathbb{R}^n, \tilde{r} \in \mathbb{R}$) as the residual vector for the nonlinear equation

$$\begin{bmatrix} s \\ \kappa \end{bmatrix} - \psi(x, \tau) = 0$$

from HLCP (6). For example, given an initial point $x^0 > 0, s^0 > 0, \tau^0 > 0, k^0 > 0$, the initial residual vector is shown as follows,

$$\begin{aligned} r^0 &= s^0 - Mx^0 - q\tau^0, \\ \tilde{r}^0 &= \kappa^0 + (x^0)^\top Mx^0/\tau^0 + (x^0)^\top q \end{aligned}$$

Next, the following lemma defines the central path of HLCP (6) and its existence.

Lemma 7. *(see [43, Thm. 2]): Considering HLCP* (6)*:*

(i) for any $0 < \theta \leq 1$*, there exists a strictly positive point* $(x > 0, \tau > 0, s > 0, \kappa > 0)$ *such that*

$$\begin{bmatrix} s \\ \kappa \end{bmatrix} - \psi(x, \tau) = \theta \begin{bmatrix} r^0 \\ \tilde{r}^0 \end{bmatrix}. \tag{10}$$

(ii) starting from $(x^0 = e, \tau^0 = 1, s^0 = e, \kappa^0 = 1)$*, for any* $0 < \theta \leq 1$ *there is a unique strictly positive point* $(x(\theta), \tau(\theta), s(\theta), k(\theta))$ *satisfying* (10) *and*

$$\begin{bmatrix} xs \\ \tau\kappa \end{bmatrix} = \theta e. \tag{11}$$

(iii) for any $0 < \theta \leq 1$*, the solution* $(x(\theta), \tau(\theta), s(\theta), \kappa(\theta))$ *in (ii) is bounded. Therefore, the defined central path*

$$C(\theta) \triangleq \{(x, \tau, s, \kappa) : (10), (11), 0 < \theta \leq 1\} \tag{12}$$

is a continuous bounded trajectory.

(iv) when $\theta \to 0$*, any limit point* $(x(\theta), \tau(\theta), s(\theta), \kappa(\theta))$ *is a maximal complementary solution for HLCP* (6)*.*

By Lemma 7, interior point algorithms can generate iterates with a neighborhood of $C(\theta)$ and converge toward a maximal complementary solution for HLCP (6).

III. Infeasible full Newton IPM Algorithm for HLCP

The previous section primarily explains how the homogeneous formulation, HLCP (6), enables the detection of infeasibility. Remarkably, this homogeneous formulation also surprisingly allows infeasible IPMs to achieve an iteration complexity of $O(\sqrt{n})$, a property traditionally reserved for feasible IPMs.

In this section, we introduce an infeasible IPM algorithm with a full Newton step, whose iteration complexity is proved to be *data independent* (*only dimension dependent*) and *exact* with the value:

$$\mathcal{N} = \left\lceil \frac{\log(\frac{n+1}{\epsilon})}{-\log(1 - \frac{0.414213}{\sqrt{n+1}})} \right\rceil.$$

We adopt the notations for simplicity,

$$\bar{x} \triangleq \mathrm{col}(x, \tau) \in \mathbb{R}^{n+1}, \ \bar{s} \triangleq \mathrm{col}(s, \kappa) \in \mathbb{R}^{n+1}.$$

At the kth iterate $(\bar{x}^k, \bar{s}^k) > 0$, the proposed algorithm adopts the full Newton direction, which is the solution of the linearized equations

$$d_{\bar{s}} - J_\psi(\bar{x}^k) d_{\bar{x}} = -\eta \bar{r}^k \tag{13a}$$

$$\bar{x}^k \odot d_{\bar{s}} + \bar{s}^k \odot d_{\bar{x}} = \gamma \bar{\mu}^k e - \bar{x}^k \odot \bar{s}^k \tag{13b}$$

where

$$\begin{aligned} \bar{r}^k &= \bar{s}^k - \psi(\bar{x}^k), \\ \bar{\mu}^k &= \frac{(\bar{x}^k)^\top \bar{s}^k}{n+1}, \end{aligned}$$

and $\eta, \gamma \in (0, 1)$ are specified parameters by our algorithm for ensuring convergence (see the later analysis).

We first summarize our proposed infeasible full Newton IPM algorithm in Algorithm 1, wherein the choice of η, γ for convergence guarantee, the scaling and initialization strategy, and the exact number iterations are discussed in Subsections III-A, III-B and III-C, respectively.

Algorithm 1 Infeasible full Newton IPM algorithm for finding a maximal complementary solution of HLCP (6)

Input: given the data (M, q) of HLCP (6), let $\beta = 0.414213, \eta = \frac{\beta}{\sqrt{n+1}}, \gamma = 1 - \eta$, and a stopping tolerance ϵ, the required exact number of iterations is $\mathcal{N} = \left\lceil \frac{\log(\frac{n+1}{\epsilon})}{-\log(1 - \frac{0.414213}{\sqrt{n+1}})} \right\rceil$

Initialize: $\bar{x} \leftarrow e, \bar{s} \leftarrow e$, $\sigma \leftarrow \max(1, Mx + q, -x^\top Mx - x^\top q)$, $M \leftarrow \frac{1}{\sigma} M$, $q \leftarrow \frac{1}{\sigma} q$;

for $k = 1, \cdots, \mathcal{N}$ **do**

1. $\bar{r} \leftarrow \bar{s} - \psi(\bar{x})$
2. $\bar{\mu} \leftarrow \frac{\bar{x}^\top \bar{s}}{n+1}$
3. calculate $(d_{\bar{x}}, d_{\bar{s}})$ by solving (13)
4. $\bar{x} \leftarrow \bar{x} + d_{\bar{x}}$
5. $\bar{s} \leftarrow \psi(\bar{x}) + \gamma \bar{r}$

end

return $(\bar{x}, \bar{s})$;

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A. Convergence analysis

Before showing the convergence analysis, we need to prove the following lemmas.

Lemma 8. *Let $\psi(\bar{x})$ be given by* (7)*. Then, for any $\bar{x} \in \mathbb{R}^{n+1}_{++}$,*

$$\begin{aligned} \bar{x}^\top \psi(\bar{x}) &= 0, \\ \bar{x}^\top J_\psi(\bar{x}) &= -\psi(\bar{x})^\top. \end{aligned} \tag{14}$$

Proof. The proof is straightforward from the definitions $\psi(\bar{x})$ (7) and $J_\psi(\bar{x})$ (8). □

Lemma 9. *Let the direction $(d_{\bar{x}}, d_{\bar{s}})$ be obtained from* (13)*. Then, the equality*

$$d_{\bar{x}}^\top d_{\bar{s}} = d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} + \eta(1-\eta-\gamma)(\bar{x}^k)^\top \bar{s}^k$$

holds.

Furthermore, by letting $\gamma = 1-\eta$, we have

$$\left(\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}\right)^\top \left(\sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right) = d_{\bar{x}}^\top d_{\bar{s}} \geq 0 \tag{15}$$

and

$$\left\|\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}\right\|^2 + \left\|\sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right\|^2 \leq \left\|\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}} + \sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right\|^2. \tag{16}$$

Proof. Premultiplying each side of (13a) by $d_{\bar{x}}^\top$ gives

$$d_{\bar{x}}^\top d_{\bar{s}} - d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} = -\eta d_{\bar{x}}^\top \bar{r}^k \tag{17}$$

and premultiplying each side of (13a) by $(\bar{x}^k)^\top$ gives

$$(\bar{x}^k)^\top d_{\bar{s}} - (\bar{x}^k)^\top J_\psi(\bar{x}^k) d_{\bar{x}} = -\eta(\bar{x}^k)^\top \bar{r}^k. \tag{18}$$

By Lemma 8, we have $(\bar{x}^k)^\top \psi(\bar{x}^k) = 0$ and $(\bar{x}^k)^\top J_\psi(\bar{x}^k) = -\psi(\bar{x}^k)^\top$, which can reduce (18) to

$$\begin{aligned} (\bar{x}^k)^\top d_{\bar{s}} + \psi(\bar{x}^k)^\top d_{\bar{x}} &= -\eta(\bar{x}^k)^\top \bar{r}^k \\ &= -\eta(\bar{x}^k)^\top \left(\bar{s}^k - \psi(\bar{x}^k)\right) \\ &= -\eta(\bar{x}^k)^\top \bar{s}^k. \end{aligned} \tag{19}$$

Also, (17) leads to

$$\begin{aligned} d_{\bar{x}}^\top d_{\bar{s}} &= d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} - \eta d_{\bar{x}}^\top \left(\bar{s}^k - \psi(\bar{x}^k)\right) \\ &= d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} - \eta \left(d_{\bar{x}}^\top \bar{s}^k - d_{\bar{x}}^\top \psi(\bar{x}^k)\right) \\ &= d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} - \eta \left(d_{\bar{x}}^\top \bar{s}^k + (\bar{x}^k)^\top d_{\bar{s}} + \eta(\bar{x}^k)^T \bar{s}^k\right) \\ &\qquad \text{(by (19))} \\ &= d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} - \eta \left[e^\top(\gamma\bar{\mu}^k e - \bar{x}^k \bar{s}^k) + \eta(\bar{x}^k)^\top \bar{s}^k\right] \\ &\qquad \text{(by (13b))} \\ &= d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} - \eta \left[\gamma(\bar{x}^k)^\top \bar{s}^k - (\bar{x}^k)^\top \bar{s}^k + \eta(\bar{x}^k)^\top \bar{s}^k\right] \\ &= d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} + \eta\,(1-\gamma-\eta)\,(\bar{x}^k)^\top \bar{s}^k, \end{aligned}$$

which completes the first part of the proof.

Then, by letting $\gamma = 1-\eta$ and by Lemma 3 ($J_\psi(\bar{x}^k) \succeq 0$), the above equality results in

$$\begin{aligned} \left(\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}\right)^\top \left(\sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right) &= d_{\bar{x}}^\top d_{\bar{s}} \\ &= d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} \geq 0, \end{aligned}$$

which can be used to derive the inequality

$$\left\|\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}\right\|^2 + \left\|\sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right\|^2 \leq \left\|\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}} + \sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right\|^2.$$

This completes the second part of the proof. □

In our proposed Algorithm 1, Step 4 ($\bar{x} \leftarrow \bar{x} + d_{\bar{x}}$) and Step 5 ($\bar{s} \leftarrow \psi(\bar{x}) + \gamma\bar{r}$) are derived from the two different update equations for $\bar{x}$ and $\bar{s}$:

$$\bar{x}^+ \triangleq \bar{x}^k + \alpha d_{\bar{x}} > 0 \tag{20}$$

and

$$\begin{aligned} \bar{s}^+ &\triangleq \bar{s}^k + \alpha d_{\bar{s}} + \psi(\bar{x}^+) - \psi(\bar{x}^k) - \alpha J_\psi(\bar{x}^k) d_{\bar{x}} \\ &= \psi(\bar{x}^+) + \left(\bar{s}^k - \psi(\bar{x}^k)\right) + \alpha\left(d_{\bar{s}} - J_\psi(\bar{x}^k) d_{\bar{x}}\right) \\ &= \psi(\bar{x}^+) + \bar{r}^k - \alpha\eta\bar{r}^k \\ &= \psi(\bar{x}^+) + (1-\alpha\eta)\bar{r}^k \end{aligned} \tag{21}$$

when choosing the step size $\alpha = 1$ and $\eta = 1-\gamma$.

Remark 2. *Thanks to this special update* (21) *for $\bar{s}$, which was also suggested in [46] for the feasible case, the next iterate of the residual vector $\bar{r}^+$ satisfies*

$$\begin{aligned} \bar{r}^+ &\triangleq \bar{s}^+ - \psi(\bar{x}^+) \\ &= (1-\alpha\eta)\bar{r}^k. \end{aligned} \tag{22}$$

Then, we have the following Lemma.

Lemma 10. *If the next iterate $(\bar{x}^+, \bar{s}^+)$ is given by (20) and (21), then*

$$(\bar{x}^+)^\top \bar{s}^+ = (1-\alpha(1-\gamma))\,(\bar{x}^k)^\top \bar{s}^k + \alpha^2\eta(1-\eta-\gamma)(\bar{x}^k)^\top \bar{s}^k.$$

Furthermore, by letting $\eta = 1-\gamma$, the residual $\bar{r}$ (infeasibility measure) and the complementarity gap $(\bar{x})^\top \bar{s}$ are reduced at the same rate $(1-\alpha\eta)$.

Proof. Based on the updates of $\bar{x}$ (20) and $\bar{s}$ (21), we have

$$\begin{aligned} &(\bar{x}^+)^\top \bar{s}^+ \\ &= (\bar{x}^+)^\top \left(\bar{s}^k + \alpha d_{\bar{s}} + \psi(\bar{x}^+) - \psi(\bar{x}^k) - \alpha J_\psi(\bar{x}^k) d_{\bar{x}}\right) \\ &= (\bar{x}^+)^\top \left(\bar{s}^k + \alpha d_{\bar{s}}\right) - (\bar{x}^k + \alpha d_{\bar{x}})^\top \left(\psi(\bar{x}^k) + \alpha J_\psi(\bar{x}^k) d_{\bar{x}}\right) \\ &= (\bar{x}^+)^\top \left(\bar{s}^k + \alpha d_{\bar{s}}\right) - (\bar{x}^k)^\top \psi(\bar{x}^k) - \alpha(d_{\bar{x}})^\top \psi(\bar{x}^k) \\ &\quad - \alpha(\bar{x}^k)^\top J_\psi(\bar{x}^k) d_{\bar{x}} - \alpha^2 d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} \\ &\qquad \text{(by Lemma 8)} \\ &= (\bar{x}^+)^\top \left(\bar{s}^k + \alpha d_{\bar{s}}\right) - \alpha^2 d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} \\ &= (\bar{x}^k + \alpha d_{\bar{x}})^\top \left(\bar{s}^k + \alpha d_{\bar{s}}\right) - \alpha^2 d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} \\ &= (\bar{x}^k)^\top \bar{s}^k + \alpha\left(d_{\bar{x}}^\top \bar{s}^k + d_{\bar{s}}^\top \bar{x}^k\right) + \alpha^2\left(d_{\bar{x}}^\top d_{\bar{s}} - d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}}\right) \\ &= (\bar{x}^k)^\top \bar{s}^k + \alpha\left(d_{\bar{x}}^\top \bar{s}^k + d_{\bar{s}}^\top \bar{x}^k\right) + \alpha^2\eta(1-\eta-\gamma)(\bar{x}^k)^\top \bar{s}^k \\ &\qquad \text{(by Lemma 9)} \\ &= (1-\alpha(1-\gamma))\,(\bar{x}^k)^\top \bar{s}^k + \alpha^2\eta(1-\eta-\gamma)(\bar{x}^k)^\top \bar{s}^k. \\ &\qquad \text{(by (13b))} \end{aligned}$$

which completed the first part of the proof.

Furthermore, by letting $\eta = 1-\gamma$, the update of the complementarity gap $\bar{x}^\top \bar{s}$ is

$$(\bar{x}^+)^\top \bar{s}^+ = (1-\alpha\eta)(\bar{x}^k)^\top \bar{s}^k, \tag{23}$$

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which is the same as the update (22) of the infeasibility residual $\bar{r}$. This completes the second part of the proof. □

Theorem 1. *Suppose that* $\|\bar{x}^k \odot \bar{s}^k - \bar{\mu}^k e\| \le \beta\bar{\mu}^k$ *where* $0 < \beta < \sqrt{2}-1$ *(e.g.,* $\beta = 0.414213$*); letting* $\eta = \frac{\beta}{\sqrt{n+1}}$, $\gamma = 1-\eta$*, and adopting the full Newton step (*$\alpha = 1$*), then* $\bar{x}^+ > 0$.

Proof. Since $\|\bar{x}^k \odot \bar{s}^k - \bar{\mu}^k e\| \le \beta\bar{\mu}^k$ where $0 < \beta < \sqrt{2}-1$, we have that

$$\|\bar{x}^k \odot \bar{s}^k - \bar{\mu}^k e\|_\infty \le \|\bar{x}^k \odot \bar{s}^k - \bar{\mu}^k e\| \le \beta\bar{\mu}^k,$$

that is,

$$-\beta\bar{\mu}^k e \le \bar{x}^k \odot \bar{s}^k - \bar{\mu}^k e \le \beta\bar{\mu}^k e$$

Thus, we have

$$\min(\bar{x}^k \odot \bar{s}^k) \ge (1-\beta)\bar{\mu}^k. \tag{24}$$

As $\alpha = 1$ and by Lemma 10, specifically (23), we have

$$\bar{\mu}^+ = \gamma\bar{\mu}^k, \tag{25}$$

where $\bar{\mu}^+ = \frac{(\bar{x}^+)^\top \bar{s}^+}{n+1}$. Therefore, (13b) is equivalent to the equality

$$\bar{s}^k \odot d_{\bar{x}} + \bar{x}^k \odot d_{\bar{s}} = \bar{\mu}^+ e - \bar{x}^k \odot \bar{s}^k,$$

which can also be reformulated as

$$\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}} + \sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}} = -\frac{\bar{x}^k \odot \bar{s}^k - \bar{\mu}^+ e}{\sqrt{\bar{x}^k \odot \bar{s}^k}}. \tag{26}$$

As a result,

$$\begin{aligned}
&\left\|\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}} + \sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right\|^2 = \left\|\frac{\bar{x}^k \odot \bar{s}^k - \bar{\mu}^+ e}{\sqrt{\bar{x}^k \odot \bar{s}^k}}\right\|^2 \\
&\le \frac{\|\bar{x}^k \odot \bar{s}^k - \bar{\mu}^+ e\|^2}{\min\left(\sqrt{\bar{x}^k \odot \bar{s}^k}\right)^2} \le \frac{\|\bar{x}^k \odot \bar{s}^k - \bar{\mu}^+ e\|^2}{(1-\beta)\bar{\mu}^k} \quad \text{(by (24))} \\
&= \frac{\|\bar{x}^k \odot \bar{s}^k - \bar{\mu}^k e + (1-\gamma)\bar{\mu}^k e\|^2}{(1-\beta)\bar{\mu}^k} \quad \text{(by (25))} \\
&= \frac{\|\bar{x}^k \odot \bar{s}^k - \bar{\mu}^k e\|^2 + \|(1-\gamma)\bar{\mu}^k e\|^2}{(1-\beta)\bar{\mu}^k} \\
&\quad \text{(as } e^\top(\bar{x}^k\bar{s}^k - \bar{\mu}^k e) = 0) \\
&\le \frac{\beta^2(\bar{\mu}^k)^2 + \eta^2(n+1)(\bar{\mu}^k)^2}{(1-\beta)\bar{\mu}^k} \text{(as } \eta = \frac{\beta}{\sqrt{n+1}}) \\
&= \frac{2\beta^2(\bar{\mu}^k)^2}{(1-\beta)\bar{\mu}^k} = \frac{2\beta^2\bar{\mu}^k}{1-\beta}.
\end{aligned} \tag{27}$$

Therefore, by Lemma 9 (namely (16)), we have that

$$\left\|\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}\right\|^2 + \left\|\sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right\|^2 \le \frac{2\beta^2\bar{\mu}^k}{1-\beta}.$$

Then we have that

$$\begin{aligned}
\left\|\frac{d_{\bar{x}}}{\bar{x}^k}\right\| &= \left\|\frac{1}{\sqrt{\bar{x}^k \odot \bar{s}^k}}\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}\right\| \\
&\le \frac{\left\|\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}\right\|}{\min\left(\sqrt{\bar{x}^k \odot \bar{s}^k}\right)} \le \frac{\sqrt{\left\|\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}\right\|^2 + \left\|\sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right\|^2}}{\min\left(\sqrt{\bar{x}^k \odot \bar{s}^k}\right)} \\
&\le \frac{\sqrt{\frac{2\beta^2\bar{\mu}^k}{1-\beta}}}{\sqrt{(1-\beta)\bar{\mu}^k}} \qquad \text{(by (24)))} \\
&= \frac{\sqrt{2}\beta}{1-\beta} < 1 \qquad \text{(as } 0 < \beta < \sqrt{2}-1)
\end{aligned}$$

Therefore,

$$\left\|\frac{d_{\bar{x}}}{\bar{x}^k}\right\|_\infty < 1,$$

which proves $\bar{x}^+ = \bar{x}^k + d_{\bar{x}} > 0$. The proof is complete. □

Theorem 2. *Suppose that* $\|\bar{x}^k \odot \bar{s}^k - \bar{\mu}^k e\| \le \beta\bar{\mu}^k$ *where* $0 < \beta < \sqrt{2}-1$ *(e.g.,* $\beta = 0.414213$*); letting* $\eta = \frac{\beta}{\sqrt{n+1}}$, $\gamma = 1-\eta$*, and adopting the full Newton step (*$\alpha = 1$*), then*

$$\|\bar{x}^+ \odot \bar{s}^+ - \bar{\mu}^+ e\| \le \beta\bar{\mu}^+,$$

where $\bar{\mu}^+ = \frac{(\bar{x}^+)^\top \bar{s}^+}{n+1}$*, and* $\bar{s}^+ > 0$.

Proof. By choosing $\alpha = 1$ and according to the update of $\bar{s}$ (21), we have

$$\begin{aligned}
&\bar{x}^+ \odot \bar{s}^+ - \bar{\mu}^+ e \\
&= \bar{x}^+ \odot \left(\bar{s}^k + d_{\bar{s}} + \psi(\bar{x}^+) - \psi(\bar{x}^k) - J_\psi(\bar{x}^k)d_{\bar{x}}\right) - \bar{\mu}^+ e \\
&= (\bar{x}^k + d_{\bar{x}}) \odot (\bar{s}^k + d_{\bar{s}}) - \bar{\mu}^+ e \\
&\quad + \bar{x}^+ \odot \left(\psi(\bar{x}^+) - \psi(\bar{x}^k) - J_\psi(\bar{x}^k)d_{\bar{x}}\right) \\
&= d_{\bar{x}} \odot d_{\bar{s}} + \bar{x}^+ \odot \left(\psi(\bar{x}^+) - \psi(\bar{x}^k) - J_\psi(\bar{x}^k)d_{\bar{x}}\right). \\
&\qquad \text{(by (13b) and (25))}
\end{aligned}$$

According to the definition of ψ and J_ψ (see (7) and (8), respectively), and the full Newton update of $\bar{x}^+ = \bar{x}^k + d_{\bar{x}}$, we have that

$$\begin{aligned}
&\psi(\bar{x}^+) - \psi(\bar{x}^k) - J_\psi(\bar{x}^k)d_{\bar{x}} \\
&= \begin{bmatrix} Mx^+ + q\tau^+ \\ \frac{-(x^+)^\top Mx^+}{\tau^+} - q^\top x^+ \end{bmatrix} - \begin{bmatrix} Mx^k + q\tau^k \\ \frac{-(x^k)^\top Mx^k}{\tau^k} - q^\top x^k \end{bmatrix} \\
&\quad - \begin{bmatrix} M & q \\ -\frac{(x^k)^\top(M+M^\top)}{\tau^k} - q^\top & \frac{(x^k)^\top Mx^k}{(\tau^k)^2} \end{bmatrix} \begin{bmatrix} d_x \\ d_\tau \end{bmatrix} \\
&= \begin{bmatrix} Md_x + qd_\tau \\ \frac{(x^k)^\top Mx^k}{\tau^k} - \frac{(x^+)^\top Mx^+}{\tau^+} - q^\top d_x \end{bmatrix} \\
&\quad - \begin{bmatrix} Md_x + qd_\tau \\ -\frac{(x^k)^\top(M+M^\top)d_x}{\tau^k} - q^\top d_x + \frac{d_\tau(x^k)^\top Mx^k}{(\tau^k)^2} \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ \frac{(x^k)^\top Mx^k + (x^k)^\top(M+M^\top)d_x}{\tau^k} - \frac{(x^+)^\top Mx^+}{\tau^+} - \frac{d_\tau(x^k)^\top Mx^k}{(\tau^k)^2} \end{bmatrix} \\
&\quad \text{(as } d_x^\top(M+M^\top)x = 2d_x^\top Mx) \\
&= \begin{bmatrix} 0 \\ \frac{d_\tau(x^k)^\top Mx^k + 2d_\tau d_x^\top Mx^k - \tau^k d_x^\top Md_x}{\tau^k(\tau^k + d_\tau)} - \frac{d_\tau(x^k)^\top Mx^k}{(\tau^k)^2} \end{bmatrix}
\end{aligned}$$

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and, by multiplying $\bar{x}^+$, we have that

$$\begin{aligned}
&\bar{x}^+ \odot \left(\psi(\bar{x}^+) - \psi(\bar{x}^k) - J_\psi(\bar{x}^k) d_{\bar{x}}\right) \\
&= \begin{bmatrix} x^k + d_x \\ \tau^k + d_\tau \end{bmatrix} \\
&\quad \odot \begin{bmatrix} 0 \\ \frac{d_\tau (x^k)^\top M x^k + 2 d_\tau d_x^\top M x^k - \tau^k d_x^\top M d_x}{\tau^k(\tau^k + d_\tau)} - \frac{d_\tau (x^k)^\top M x^k}{(\tau^k)^2} \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ \frac{d_\tau (x^k)^\top M x^k + 2 d_\tau d_x^\top M x^k - \tau^k d_x^\top M d_x}{\tau^k} \end{bmatrix} \\
&\quad - \begin{bmatrix} 0 \\ \frac{d_\tau (\tau^k + d_\tau)(x^k)^\top M x^k}{(\tau^k)^2} \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ \frac{\tau^k d_\tau (x^k)^\top M x^k + 2\tau^k d_\tau d_x^\top M x^k - (\tau^k)^2 d_x^\top M d_x}{(\tau^k)^2} \end{bmatrix} \\
&\quad - \begin{bmatrix} 0 \\ \frac{(\tau^k d_\tau + d_\tau^2)(x^k)^\top M x^k}{(\tau^k)^2} \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ \frac{2\tau^k d_\tau d_x^\top M x^k - (\tau^k)^2 d_x^\top M d_x - d_\tau^2 (x^k)^\top M x^k}{(\tau^k)^2} \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ -\left(d_x^\top M d_x - 2\frac{d_\tau}{\tau^k} d_x^\top M x^k + \frac{d_\tau^2}{(\tau^k)^2}(x^k)^\top M x^k\right) \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ -\left(d_x - \frac{d_\tau}{\tau^k} x^k\right)^\top M \left(d_x - \frac{d_\tau}{\tau^k} x^k\right) \end{bmatrix}.
\end{aligned}$$

By the positive semi-definiteness of M and Lemmas 3 and 9, we have that

$$\begin{aligned}
&\left\|\bar{x}^+ \odot \left(\psi(\bar{x}^+) - \psi(\bar{x}^k) - J_\psi(\bar{x}^k) d_{\bar{x}}\right)\right\| \\
&= \left(d_x - \frac{d_\tau}{\tau^k} x^k\right)^\top M \left(d_x - \frac{d_\tau}{\tau^k} x^k\right) \\
&= d_{\bar{x}}^\top J_\psi(\bar{x}^k) d_{\bar{x}} = d_{\bar{x}}^\top d_{\bar{s}}.
\end{aligned}$$

Then, by summarizing the above analysis, we have that

$$\begin{aligned}
&\|\bar{x}^+ \odot \bar{s}^+ - \bar{\mu}^+ e\| \\
&\le \|d_{\bar{x}} \odot d_{\bar{s}}|\| + \left\|\bar{x}^+ \odot \left(\psi(\bar{x}^+) - \psi(\bar{x}^k) - J_\psi(\bar{x}^k) d_{\bar{x}}\right)\right\| \\
&\le \|d_{\bar{x}} \odot d_{\bar{s}}\| + d_{\bar{x}}^T d_{\bar{s}} = \left\|\left(\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}\right) \odot \left(\sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right)\right\| \\
&\quad + \left(\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}\right)^\top \sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}.
\end{aligned}$$

For simplicity, we introduce two vectors $a \triangleq \sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}}$ and $b \triangleq \sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}$; then we have that

$$\begin{aligned}
&\|\bar{x}^+ \odot \bar{s}^+ - \bar{\mu}^+ e\| \le \|a \odot b\| + a^\top b \le \|a\| \cdot \|b\| + a^\top b \\
&= \|a\| \cdot \|b\| + \tfrac{1}{2}\|a+b\|^2 - \tfrac{1}{2}\left(\|a\|^2 + \|b\|^2\right) \\
&= \tfrac{1}{2}\|a+b\|^2 - \tfrac{1}{2}(\|a\| - \|b\|)^2 \\
&\le \tfrac{1}{2}\|a+b\|^2 = \tfrac{1}{2}\left\|\sqrt{\frac{\bar{s}^k}{\bar{x}^k}} \odot d_{\bar{x}} + \sqrt{\frac{\bar{x}^k}{\bar{s}^k}} \odot d_{\bar{s}}\right\|^2 \\
&\le \frac{\beta^2 \bar{\mu}^k}{1-\beta} = \frac{\beta}{(1-\beta)\gamma}(\beta\bar{\mu}^+).
\end{aligned}$$

by (27) and (25)

Therefore, to prove $\|\bar{x}^+\bar{s}^+ - \bar{\mu}^+ e\| \le \beta\bar{\mu}^+$, we only need to prove the inequality

$$\frac{\beta}{(1-\beta)\gamma} \le 1,$$

which is equivalent to proving that

$$\frac{\beta}{1-\beta} \le 1 - \frac{\beta}{\sqrt{n+1}}, \quad \forall n \ge 1,$$

as $\gamma = 1 - \eta = 1 - \frac{\beta}{\sqrt{n+1}}$. When $0 < \beta < \sqrt{2} - 1$, $\forall n \ge 1$, the inequalities

$$\frac{\beta}{1-\beta} + \frac{\beta}{\sqrt{n+1}} \le \frac{\beta}{1-\beta} + \frac{\beta}{\sqrt{2}} < \frac{\sqrt{2}-1}{2-\sqrt{2}} + \frac{\sqrt{2}-1}{\sqrt{2}} = 1$$

hold, which completes the first part of the proof.

Now we have that $\|\bar{x}^+ \odot \bar{s}^+ - \bar{\mu}^+ e\| \le \beta\bar{\mu}^+$; thus

$$\|\bar{x}^+ \odot \bar{s}^+ - \bar{\mu}^+ e\|_\infty \le \|\bar{x}^+ \odot \bar{s}^+ - \bar{\mu}^+ e\| \le \beta\bar{\mu}^+.$$

That is,

$$-\beta\bar{\mu}^+ e \le \bar{x}^+ \odot \bar{s}^+ - \bar{\mu}^+ e \le \beta\bar{\mu}^+ e,$$

which clearly shows that $\min(\bar{x}^+ \odot \bar{s}^+) \ge (1-\beta)\bar{\mu}^+ > 0$.

By Lemma 1 we have $\bar{x}^+ > 0$, so we have $\bar{s}^+ > 0$, which completes the second part of the proof. □

Summarizing Lemmas 1 and 2, if the current iterate $(\bar{x}^k, \bar{s}^k)$ satisfies $\|\bar{x}^k\bar{s}^k - \bar{\mu}^k e\| \le \beta\bar{\mu}^k$ (i.e., is located in a neighborhood of the central path), letting $\beta = 0.414213$, $\eta = \frac{1}{\sqrt{n+1}}$, $\gamma = 1 - \eta$, $\alpha = 1$, and adopting the update for $\bar{x}$ (20) and $\bar{s}$ (21). The next iterate $(\bar{x}^+, \bar{s}^+)$ will keep positive and stay in the neighborhood of the central path. Clearly, the initialization strategy (29) ensures that

$$\|\bar{x}^0 \odot \bar{s}^0 - \bar{\mu}^0\| = 0 \le \beta\bar{\mu}^0, \quad \bar{x}^0 > 0, \quad \bar{s}^0 > 0,$$

so the iterates, $(\bar{x}^k, \bar{s}^k)$ $\forall k = 1, 2, \cdots$ generated by Algorithm 1 satisfy

$$\begin{aligned}
&\|\bar{x}^k \odot \bar{s}^k - \bar{\mu}^k\| \le \beta\bar{\mu}^k, \\
&\bar{x}^k > 0, \quad \bar{s}^k > 0.
\end{aligned}$$

Moreover, by Lemma 10, the infeasibility residual $\bar{r}$ and the complementarity gap $\bar{x}^\top\bar{s}$ are both reduced at the same rate $(1-\eta) = 1 - \frac{0.414213}{\sqrt{n+1}}$. Therefore, by Lemma 7, the iterates generated by Algorithm 1 will converge to a maximal complementary solution for the HLCP.

The number of iterations $\mathcal{N}$ required by Algorithm 1 will depend on the initial infeasibility residual $\bar{r}^0$ and the complementarity gap $(\bar{x}^0)^\top\bar{s}^0$, which is discussed in the next Subsection.

B. Scaling and initialization strategy

To develop a QP algorithm with *data-independent* (*only dimension-dependent*) iteration complexity, we first introduce a scaling strategy for the matrix M and the vector q in the LCP as

$$\bar{M} \leftarrow \frac{1}{\sigma} M, \quad \bar{q} \leftarrow \frac{1}{\sigma} q, \tag{28}$$

where

$$\sigma = \max(1, Me + q, -e^\top M e - e^\top q).$$

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This scaling strategy (28) does not affect the optimal value of x in (LCP).

After that, we adopt the initialization strategy

$$x^0 = e, \quad \tau^0 = 1, \quad s^0 = e, \quad \kappa^0 = 1. \tag{29}$$

Lemma 11. *The scaling strategy (28) and the initialization strategy (29) ensure that the initial complementarity (duality) gap*

$$(\bar{x}^0)^\top \bar{s}^0 = (x^0)^\top s^0 + (\tau^0)^\top \kappa^0 = n+1$$

and the initial infeasibility residual norm

$$\|\bar{r}^0\| \le n+1.$$

Proof. Clearly, the initialization strategy (29) results in the initial complementarity gap

$$(\bar{x}^0)^\top \bar{s}^0 = (x^0)^\top s^0 + (\tau^0)^\top \kappa^0 = n+1.$$

After adopting the scaling strategy (28), the initial infeasibility residual

$$\bar{r}^0 = \begin{bmatrix} e - \bar{M}e - \bar{q} \\ 1 + e^\top \bar{M}e + e^\top \bar{q} \end{bmatrix}$$

and the scaling factor σ ensure that all elements of $\bar{r}^0$ are nonnegative (i.e., $\bar{r}^0 \ge 0$). Thus, we have that

$$\|\bar{r}^0\|^2 \le (e^\top \bar{r}^0)^2 = (n+1)^2$$

that is, $\|\bar{r}^0\| \le n+1$. This completes the proof. □

Thus, our proposed scaling and initialization strategies make the initial complementarity gap $(\bar{x}^0)^\top \bar{s}^0$ and initial infeasibility residual $\bar{r}^0$ *independent* from the problem data (M, q) (*only dependent on the problem dimension* n). For some warm-start initialization strategies, such as the previous solution in MPC, it is very challenging to derive an **data-independent** upper bound for the initial complementarity (dual) gap $(\bar{x}^0)^\top \bar{s}^0$ and the initial infeasibility residual norm $\|\bar{r}^0\|$, unlike Eqn. (29).

C. Exact iteration complexity

Thanks to the above scaling and initialization strategies, our proposed infeasible full Newton IPM Algorithm 1 is *only dimension dependent* (*data independent*), is *simple to calculate*, and has *exact* iteration complexity.

Theorem 3. *The infeasible full Newton IPM Algorithm 1 returns a ϵ-approximate maximal complementary solution of HLCP (6) in an exact*

$$\mathcal{N} = \left\lceil \frac{\log(\frac{n+1}{\epsilon})}{-\log(1 - \frac{0.414213}{\sqrt{n+1}})} \right\rceil$$

iterations, where an ϵ-approximate maximal complementary solution satisfies $\|\bar{r}\| \le \epsilon,\ \bar{x}^\top \bar{s} \le \epsilon$.

Proof. Since our previous summarization of Theorem 1 and Theorem 2 (see the last paragraph of the Subsection III-A), shows the convergence of Algorithm 1, here we only show how to derive the required number of iterations for satisfying $\|\bar{r}^k\| \le \epsilon$ and $(\bar{x}^k)^\top \bar{s}^k \le \epsilon$.

By Lemma 10, the infeasibility residual $\bar{r}^k$ and complementarity gap $(\bar{x}^k)^\top \bar{s}^k$ at the kth iterate are

$$\begin{aligned} \bar{r}^k &= (1-\eta)^k \bar{r}^0, \\ (\bar{x}^k)^\top \bar{s}^k &= (1-\eta)^k (\bar{x}^0)^\top \bar{s}^0. \end{aligned} \tag{30}$$

Then, the inequalities $\|\bar{r}^k\| \le \epsilon$ and $(\bar{x}^k)^\top \bar{s}^k \le \epsilon$ hold if

$$\begin{aligned} (1-\eta)^k \|\bar{r}^0\| &\le \epsilon, \\ (1-\eta)^k (\bar{x}^0)^\top \bar{s}^0 &\le \epsilon. \end{aligned}$$

Taking logarithms gives

$$\begin{aligned} k\log(1-\eta) + \log(\|\bar{r}^0\|) &\le \log(\epsilon), \\ k\log(1-\eta) + \log((\bar{x}^0)^\top \bar{s}^0) &\le \log(\epsilon). \end{aligned}$$

Thus, the inequalities $\|\bar{r}^k\| \le \epsilon$ and $(\bar{x}^k)^\top \bar{s}^k \le \epsilon$ hold if

$$k \ge \left\lceil \frac{\log(\frac{\max(\|\bar{r}^0\|, (\bar{x}^0)^\top \bar{s}^0)}{\epsilon})}{-\log(1-\eta)} \right\rceil = \left\lceil \frac{\log(\frac{n+1}{\epsilon})}{-\log(1-\eta)} \right\rceil.$$

(by Lemma 11, $\max(\|\bar{r}^0\|, (\bar{x}^0)^\top \bar{s}^0) = n+1$)

The expressions (30) hold as equalities rather than inequalities, thus the number of iterations is *exact*, rather than an upper bound. Therefore, after the exact number of iterations

$$\mathcal{N} = \left\lceil \frac{\log(\frac{n+1}{\epsilon})}{-\log(1 - \frac{0.414213}{\sqrt{n+1}})} \right\rceil, \tag{31}$$

the inequalities $\|\bar{r}^k\| \le \epsilon$ and $(\bar{x}^k)^\top \bar{s}^k \le \epsilon$ hold, which completes the proof. □

IV. Algorithm implementation

A. Robust procedure for solving Newton systems

In Algorithm 1, the most computationally expensive step is Step 3, solving the linear system (13). To meet our mentioned execution time certificate requirement, we cannot adopt iterative methods to solve the linear system (13) as their [flops] are hard to determine and *data dependent*; here we have to use direct methods (such as LU, QR, and Cholesky decomposition) as their [flops] are determined and *data independent*.

In fact, Algorithm 1 does not use $d_{\bar{s}}$ (as $\bar{s} \leftarrow \psi(\bar{x}) + \gamma\bar{r}$, rather than $\bar{s} \leftarrow \bar{s} + d_{\bar{s}}$). By eliminating $d_{\bar{s}}$ with

$$d_{\bar{s}} = \frac{\bar{s}^k}{\bar{x}^k} d_{\bar{x}^k} + \gamma\bar{\mu}^k \frac{1}{\bar{x}^k} - \bar{s}^k,$$

we can reduce the linear system (13) to the compact system

$$\left(J_\psi(\bar{x}^k) + \operatorname{diag}\left(\frac{\bar{s}^k}{\bar{x}^k}\right) \right) d_{\bar{x}} = \gamma\bar{\mu}^k \frac{1}{\bar{x}^k} - \bar{s}^k + \eta\bar{r}^k. \tag{32}$$

Remark 3. *By Lemma 3, $J_\psi(\bar{x}^k)$ is positive semi-definite. Thus, for arbitrary $\bar{x}^k, \bar{s}^k \in \mathbb{R}^{n+1}_{++}$, (32) is a positive definite system and thus there exists a unique solution.*

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After adopting the scaling (28) and according to the definition J_ψ (8), M, and $\bar{x} = \mathrm{col}(z, y, \tau), \bar{s} = \mathrm{col}(v, w, \kappa)$, the linear system (32) is expanded as

$$\begin{bmatrix} Q(t) + \mathrm{diag}(\frac{v^k}{z^k}) & -A(t)^\top & c(t) \\ A(t) & \mathrm{diag}(\frac{w^k}{y^k}) & -b(t) \\ -\frac{2(z^k)^\top Q(t)}{\tau^k} - c(t)^\top & b(t)^\top & \frac{(z^k)^\top Q(t) z^k}{(\tau^k)^2} + \frac{\kappa^k}{\tau^k} \end{bmatrix} \cdot \begin{bmatrix} d_z \\ d_y \\ d_\tau \end{bmatrix} = \begin{bmatrix} \gamma\bar{\mu}\frac{1}{z^k} - v^k + \eta r_z \\ \gamma\bar{\mu}\frac{1}{y^k} - w^k + \eta r_y \\ \gamma\bar{\mu}\frac{1}{\tau^k} - \kappa^k + \eta r_\tau \end{bmatrix} \tag{33}$$

where $r_z \triangleq v^k - Q(t)z^k + A(t)^\top y^k - c(t)\tau^k$, $r_y \triangleq w^k - A(t)z^k + b(t)\tau^k$, $r_\tau \triangleq \kappa^k + (z^k)^\top Q(t) z^k + c(t)^\top z^k - b(t)^\top y^k$, and $\bar{\mu} \triangleq \frac{(z^k)^\top v^k + y^k)^\top w^k + \tau^k \kappa^k}{n+1}$.

The linear system (33) is not a symmetric positive definite system, and we tried to reduce it to a smaller symmetric positive definite system to allow the use of an efficient Cholesky decomposition method, but this approach is not robust and cannot provide a high-accuracy solution when close to the end of the iterations. This occurs because, when close to the end of the iterations, the linear system (33) is too ill-conditioned, which is the cost of the use of the homogeneous formulation. For solving (33), we choose to call Lapack's function *dgesv* [47], which uses the LU decomposition with partial pivoting and row interchanges.

Future work will investigate an effective preconditioner to allow the use of Cholesky decomposition (which has half of the cost of LU factorization) on the smaller compact system. Currently, we observe that the diagonal preconditioner reduces the condition number of (33).

B. Infeasibility reporting criteria

As for the iterates $\{x^k, \tau^k, s^k, \kappa^k\}$ generated by Algorithm 1, Lemma 6 indicates that τ^k will be bounded away from zero and κ^k will quadratically converge to zero if the QP (1) is feasible, or κ^k will be bounded away from zero and τ^k will quadratically converge to zero if the QP (1) is infeasible. Thus, this paper adopts $\tau < \kappa$ as the infeasibility reporting criteria.

C. Execution time certificate

Summarizing the above analysis, we rewrite Algorithm 1 as an implementable Algorithm 2. Note that Step 1 of Algorithm 1 is equal to Step 5 (before the iteration) and Step 6.6 of Algorithm 2. In addition to using the Lapack library, we also use the efficient BLAS library [48] to implement matrix-vector multiplications and vector inner product operations of Algorithm 2.

Thanks to the *exact* iteration complexity and the use of LU decomposition in solving the linear system (33), the total [flops] of Algorithm 2 is easily countable and fixed. In practice, running Algorithm 2 once on a given processor, the execution time is obtained and will almost surely be the same among all QPs with the same problem dimension.

Algorithm 2 EIQP: execution-time-certified and infeasibility-detecting QP solver, an infeasible homogeneous full Newton IPM algorithm for convex QP (1).

Input: given the data of QP (1) $(Q(t), c(t), A(t), b(t))$, $\beta = 0.414213, \eta = \frac{\beta}{\sqrt{n+1}}, \gamma = 1 - \eta$, and a optimality level ϵ, and then the required exact number of iterations is $\mathcal{N} = \left\lceil \frac{\log(\frac{n+1}{\epsilon})}{-\log(1 - \frac{0.414213}{\sqrt{n+1}})} \right\rceil$

1. $z \leftarrow e,\ y \leftarrow e,\ \tau \leftarrow 1,\ v \leftarrow e,\ w \leftarrow e,\ \kappa \leftarrow 1$;
2. cache: $Qz \leftarrow Q(t)z, Ay \leftarrow A(t)^\top y, Az \leftarrow A(t)z, zc \leftarrow z^\top c(t), yb \leftarrow y^\top b(t), zQz \leftarrow z^\top Qz$;
3. $\sigma \leftarrow \max(1, Qz - Ay + c(t), Az - b(t), -zQz - zc + yb)$;
4. scale: $Q(t) \leftarrow \frac{1}{\sigma}Q(t), c(t) \leftarrow \frac{1}{\sigma}c(t), A(t) \leftarrow \frac{1}{\sigma}A(t), b(t) \leftarrow \frac{1}{\sigma}b(t), Qz \leftarrow \frac{1}{\sigma}Qz, Ay \leftarrow \frac{1}{\sigma}Ay, Az \leftarrow \frac{1}{\sigma}Az, zc \leftarrow \frac{1}{\sigma}zc, yb \leftarrow \frac{1}{\sigma}yb, zQz \leftarrow \frac{1}{\sigma}zQz$;
5. $r_z \leftarrow v - Qz + Ay - c(t)\tau,\ r_y \leftarrow w - Az + b(t)\tau,\ r_\tau \leftarrow \kappa + \frac{zQz}{\tau} + zc - yb$;
6. **for** $k = 1, \cdots, \mathcal{N}$ **do**
   6.1. $\bar{\mu} \leftarrow \frac{z^\top v + y^\top w + \tau\kappa}{n+1}$;
   6.2. call *dgesv* to solve (33), obtain d_z, d_y, d_τ;
   6.3. $z \leftarrow z + d_z,\ y \leftarrow y + d_y,\ \tau \leftarrow \tau + d_\tau$;
   6.4. cache: $Qz \leftarrow Q(t)z, Ay \leftarrow A(t)^\top y, Az \leftarrow A(t)z, zc \leftarrow z^\top c(t), yb \leftarrow y^\top b(t), zQz \leftarrow z^\top Qz$;
   6.5. $v \leftarrow Qz - Ay + c(t)\tau + \gamma r_z,\ w \leftarrow Az - b(t)\tau + \gamma r_y,\ \kappa \leftarrow -\frac{zQz}{\tau} - zc + yb + \gamma\tilde{r}$;
   6.6. $r_z \leftarrow \gamma r_z,\ r_y \leftarrow \gamma r_y, r_\tau \leftarrow \gamma r_\tau$;
   end
7. **if** $\tau < \kappa$, **return** (1) is infeasible; **otherwise**,
8. **return** $z = \frac{1}{\tau}z$.

D. C-code implementation

The C-code implementation of Algorithm 2 is just one single C file, making it easy to integrate with other software and appealing in deployment in embedded production systems. Its MATLAB, Julia, and Python interfaces and numerical examples are publicly available at https://github.com/liangwu2019/EIQP.

The following reported simulation results were obtained on a Mac mini with an Apple M4 Chip (10-core CPU and 16 GB RAM).

V. Numerical Experiments

This section first verifies whether Algorithm 2 exhibits the same iteration complexity in practice as stated in Theorem 3. It

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then compares the infeasibility-detection capabilities of Algorithm 2 with the known MATLAB's "Quadprog" solver and several popular open source QP solvers, including "OSQP" [22], "SCSv3" [49], and "Clarabel" [44] by running on random infeasible QP examples. It finally compares the computation speed and solution accuracy of Algorithm 2 with "Quadprog", "OSQP", "SCSv3", and "Clarabel" on highly ill-conditioned feasible QP problems.. Because in such real-time RTI-based NMPC scenarios, the resulting condensing QPs with time-varying data are usually ill-conditioned.

A. Validation of number of iterations

To validate Theorem 3, we verify whether Algorithm 2 can achieve both the duality gap and the infeasibility residual norm below the specified optimality level ϵ within exactly $\left\lceil \frac{\log\left(\frac{n+1}{\epsilon}\right)}{-\log\left(1-\frac{0.414213}{\sqrt{n+1}}\right)} \right\rceil$ iterations.

We first generate random feasible QP problems with different sizes ($n = 100, 200$). Then, we examine how many iterations are required by Algorithm 2 to reduce both the duality gap and the infeasibility residual norm to the given optimality level $\epsilon = 10^{-6}, 10^{-8}$. Fig. 1 shows how both the duality gap and the infeasibility residual norm decrease with the number of iterations. In both $n = 100$ and $n = 200$ cases, Fig. 1 shows that the duality gap and the infeasibility residual norm arrive the given optimality level $\epsilon = 10^{-6}, 10^{-8}$ in a **exact** and **smaller** number of iterations given by Eqn.(31) of Theorem 3, respectively, which is consistent with the analysis of Lemma 11: the initial duality gap $(\bar{x}^0)^\top \bar{s}^0 = n+1$ and the initial infeasibility residual norm $\|\bar{r}^0\| \leq n+1$. Therefore, after **exact** $\left\lceil \frac{\log\left(\frac{n+1}{\epsilon}\right)}{-\log\left(1-\frac{0.414213}{\sqrt{n+1}}\right)} \right\rceil$ iterations (Eqn.(31)), the duality gap and the infeasibility residual norm are both reduced to the given optimality level ϵ.

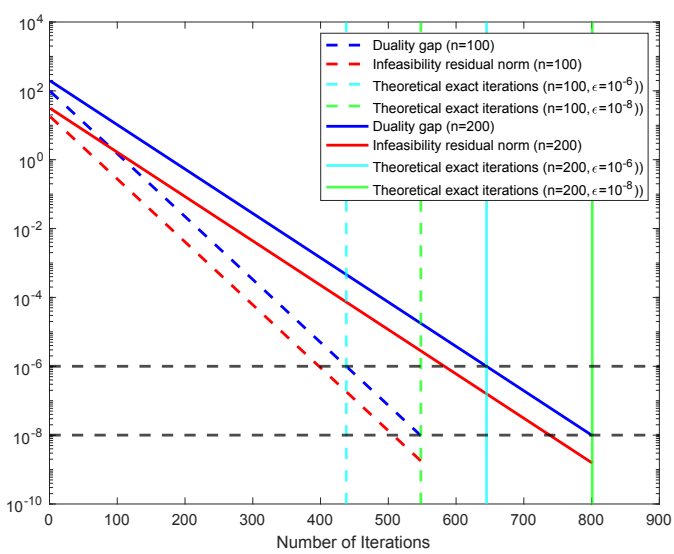

Fig. 1. Duality gap and Infeasibility residual norm vs. the number of iterations, and the theoretical exact number of iterations by applying Algorithm 2 to random feasible QP problems for $n = 100, 200$ and $\epsilon = 10^{-6}, 10^{-8}$.

B. Random infeasible QP examples

To demonstrate the infeasibility-detection capability of our proposed Algorithm 2, we randomly generate 100 QPs ($\min \frac{1}{2} z^\top Q z + c^\top z$, s.t. $Az \leq b$) for each case: condition number $cond(Q)$ varies from 10^1 to 10^6. To make QPs infeasible, we then add the contradictory constraints: $A \leftarrow [A; -A(1,:); -A(2,:)], b \leftarrow [b; -b(1)-1; -b(2)-1]$. As for solver settings, "Quadprog", "OSQP", "SCSv3", and "Clarabel" use their default settings. Our "EIQP" chooses $\epsilon = 10^{-6}$. For each condition number setting, 100 random experiments are conducted to obtain statistical results on the infeasibility detection rate. The infeasibility detection rate of these QP solvers is plotted in Fig. 2.

The "Quadprog" solver has poor infeasibility-detection results, and its detection rate gradually becomes lower as the condition number of Q increases. The "OSQP" and "SCSv3" solvers also cannot reach 100%, and it is hard to tune the solver settings of "OSQP" and "SCSv3" to achieve 100% detection rate. "Clarabel" and our EIQP solvers have a 100% infeasibility detection rate among all cases, consistent with the proof.

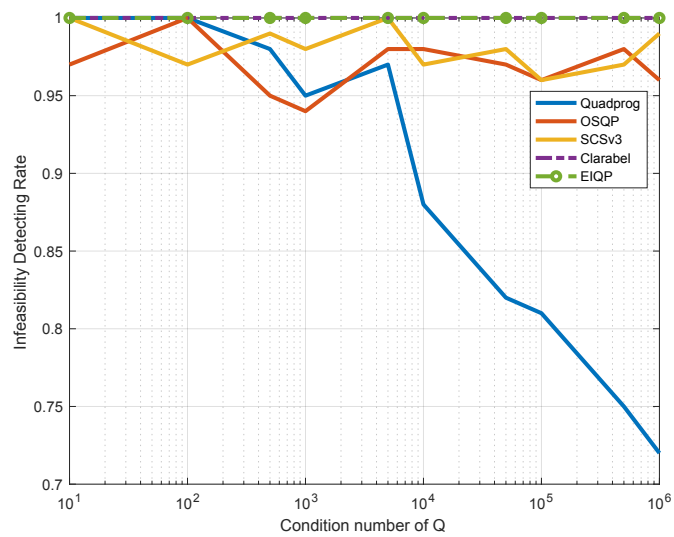

Fig. 2. Comparison of Quadprog, OSQP, SCSv3, Clarabel, and EIQP solvers in detecting infeasibility of random infeasible QP problems as a function of the condition number of the Hessian matrix Q.

C. Feasible ill-conditioned QP examples

The real-time RTI-based NMPC approach often encounters an ill-conditioning issue when formulating a long-horizon MPC into a condensed time-varying QP. Thus, it is necessary to investigate whether Algorithm 2 and "Quadprog", "OSQP", "SCSv3", and "Clarabel" solvers can handle highly ill-conditioned feasible QP problems. For a fair comparison, we first assign the optimal solution $z^* = \mathrm{col}(\mathbf{1}_{n_z/2}, \mathbf{0}_{n_z/2})$ (the relative accuracies of different solvers are based on this), and then the matrix Q is randomly generated with condition number 10^6 and matrix A is dense randomly generated with near-linear-dependence (10% percent of rows are almost the same with the first row). Extreme column scaling and row scaling (from $10^{-3} \sim 10^3$) are then applied to Q and

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A to make further KKT very ill-conditioned. By randomly choosing an active constraint set, b and c are generated by the corresponding KKT condition (if i is active, $b_i = A_{i,:}z^*$, else $b_i = A_{i,:}z^* - 0.5$; $c = A^\top y + v - Qz^*$ where the values of y and v are based on the active constraint set) to ensure that the optimal solution is $z^* = \mathrm{col}(\mathbf{1}_{n_z/2}, \mathbf{0}_{n_z/2})$. We test the problem dimension with $n_z \in [50, 500]$ and set $n_b = n_z/5$.

Fig. 3 shows the computation time and solution accuracy of different solvers. "Quadprog" (with the default setting) ranks first (best) in terms of solution accuracy among all solvers, and ranks third in computation time. "OSQP" (settings: 'max_iter',1e5,'eps_abs',1e-12,'eps_rel',1e-12,') always reaches the maximum iteration limit and exhibits severe accuracy issues (around 10^0 level), and ranks second to last in terms of computation time. Similarly, "SCSv3" (settings: 'max_iter',1e4,'eps_abs',1e-12,'eps_rel',1e-12,') reaches the maximum iteration limit and exhibits severe accuracy issues, and ranks last in terms of computation time (even up to 18 seconds on $n_z = 500$ cases). "Clarabel" (with default settings) reports the status "SOLVED" but yields the lowest solution accuracy (around the 10^0 level), so its fastest computation speed is not practically meaningful. "EIQP" (Algorithm 2 with $\epsilon = 10^{-12}$) ranks second in solution accuracy (around level 10^{-2}), but spends more computation time compared to "Quadprog", showing no advantage. As theoretical iteration bounds are often conservative, and if pursuing computational efficiency without considering the iterations certificate, we can implement a heuristic variant, "EIQP_heuristic", with $\gamma = 1 - \frac{0.414213}{\log(n)}$, inspired by [50] to achieve anticipated $O(\log(n))$ behavior (proving $O(\log(n))$ iteration complexity remains an open challenge for over four decades in IPM field). From Fig. 3, "EIQP_heuristic" attains the same solution accuracy as "EIQP" while delivering up $10\times$ speedup, ranking the fastest solvers (very close to "Clarabel").

Discussion of limitations: Our current "EIQP" does not exhibit the same robustness as "Quadprog". As one future work, we will implement a similar presolve phase as used in "Quadprog", to enhance solution accuracy for ill-conditioned QPs. Although "EIQP_heuristic" has practical computation speed, it lacks iteration complexity proof (also probably not in the future). Our upcoming version "EIQPv2", based on the adaptive-step predictor-corrector framework of [51] (not the above heuristic rule), will achieve the certified $O(\sqrt{n})$ iteration complexity and while enjoys the anticipated $O(\log(n))$ behavior, thereby delivering state-of-the-art computational speed.

VI. Control Applications

In real-time optimization-based control applications, the certified worst-case computation time is far more critical than the average experimental computation time. Having a certified relationship between execution time and problem dimension in advance enables the selection of controller complexity that meets feedback sampling time requirements. This section picks up the AFTI-16 MPC example and ACC CBF-QP example to demonstrate that our proposed Algorithm 2 (EIQP solver) can provide such an execution time certificate.

A. Real-time MPC applications

A general linear MPC for tracking is

$$\begin{aligned} \min\ & \frac{1}{2}\sum_{k=0}^{N_p-1} \|Cx_{k+1} - r(t))\|_{W_y}^2 + \|u_k - u_{k-1}\|_{W_{\Delta u}}^2 \\ \text{s.t. } & x_{k+1} = Ax_k + Bu_k,\ k = 0, \dots, N_p - 1, \\ & E_x x_{k+1} + E_u u_k \le f_k,\ k = 0, \dots, N_p - 1, \\ & x_0 = x(t), \quad u_{-1} = u(t-1), \end{aligned} \tag{34}$$

where $x_k \in \mathbb{R}^{n_x}$ denotes the states and $u_k \in \mathbb{R}^{n_u}$ denotes the control inputs. The control objective is to minimize the tracking errors between the output Cx_{k+1} and the reference signal $r(t)$, while penalizing the control input increments Δu_k, along the prediction horizon N_p. The matrices $W_y \succ 0$ and $W_{\Delta u} \succ 0$ are the weights for the output tracking error and control input increments, respectively. $x(t)$ and $u(t-1)$ denote the current feedback states and previous control inputs, respectively. $E_x x_{k+1} + E_u u_k \le f_k$ denotes the general constraints formulation such as the box constraints $u_{\min} \le u_k \le u_{\max},\ x_{\min} \le x_{k+1} \le x_{\max}$ or the terminal constraints $x_{N_p} \in \mathcal{X}_{N_p}$. Denoting

$$z \triangleq \mathrm{col}(u_0 - u_{\min}, \cdots, u_{N_p-1} - u_{\min}) \in \mathbb{R}^{n_z}$$

($n_z = N_p n_u$), we can eliminate the states (see [52]), to get a compact QP (1). Here we consider the classical AFTI-16 example from [53], [54]:

$$\begin{cases} \dot{x} = \begin{bmatrix} -0.0151 & -60.5651 & 0 & -32.174 \\ -0.0001 & -1.3411 & 0.9929 & 0 \\ 0.00018 & 43.2541 & -0.86939 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} x \\ \qquad + \begin{bmatrix} -2.516 & -13.136 \\ -0.1689 & -0.2514 \\ -17.251 & -1.5766 \\ 0 & 0 \end{bmatrix} u \\ y = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} x \end{cases}$$

which is sampled using zero-order hold every 50 ms. Our MPC controller feedback time is also 50 ms, so we need to ensure that the MPC design meets this execution time requirement. The input constraints are $|u_i| \le 25°, i = 1, 2$, the output constraints are $-0.5 \le y_1 \le 0.5$ and $-100 \le y_2 \le 100$. The control goal is to make the pitch angle y_2 track a reference signal r_2. In designing the MPC controller, we take $W_y = \mathrm{diag}([10,10])$, $W_u = 0$, and $W_{\Delta u} = \mathrm{diag}([0.1,\ 0.1])$.

Now, the remaining MPC setting is to choose the prediction horizon to meet the requirement that the execution time is smaller than the feedback time 50 ms and maintain good closed-loop performance.

We test our EIQP solver ($\epsilon = 10^{-8}$) for different prediction horizon settings: $N_p = 5, 10, 15, 20, 25$, and their execution times, MPC closed-loop cost (average), and MPC constraint violations are listed in Table I. For the prediction horizon $N_p \le 20$, the execution time of our EIQP is less than the feedback time of 50 ms. As the prediction horizon N_p increases, the MPC tracking costs decrease but the MPC gradually violates the constraints in $N_p \ge 15$ settings, this is because

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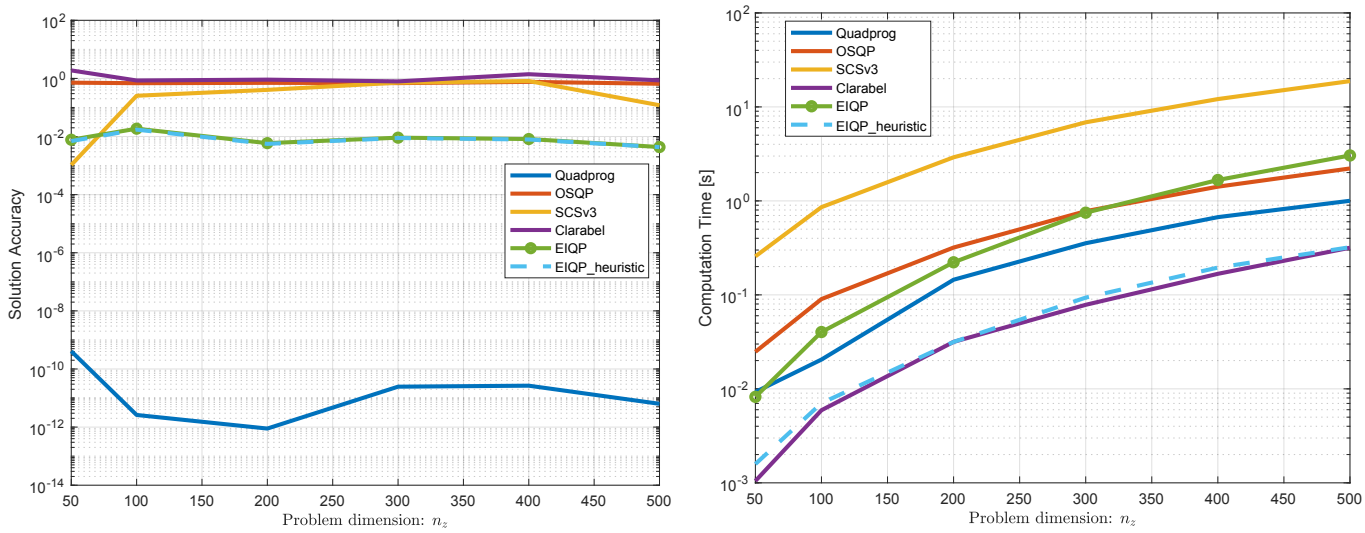

(a) (b)

Fig. 3. Performance of different solvers on QP problems with varying dimensions under extreme ill-conditioning: (a) solution accuracy vs. problem dimension n_z; (b) computation time (s) vs. problem dimension n_z.

as the prediction horizon N_p increases, the resulted QPs become gradually ill-conditioning, and thus requires smaller optimality level, which can reduce the constraint violations to 0 but increase the execution time. In our given processor, $N_p = 10$ should be chosen. The closed-loop simulation results for $N_p = 10$ are plotted in Fig. 4, which shows that the pitch angle correctly tracks the reference signal from $0°$ to $10°$ and then back to $0°$, and that both the input and output constraints are satisfied.

TABLE I
EIQP ($\epsilon = 10^{-8}$) COMPUTATIONAL PERFORMANCE FOR MPC WITH DIFFERENT PREDICTION HORIZON ($N_p = 5, 10, 15, 20, 25$) SETTINGS

Prediction horizon	Execution time [ms]	Less than 50 [ms]?	MPC costs (avg)	MPC constraint violations (avg)
$N_p = 5$	1.6	✓	42.6218	0.0
$N_p = 10$	8.7	✓	42.5558	0.0
$N_p = 15$	22.8	✓	42.5336	**0.0001**
$N_p = 20$	47.6	✓	42.4517	**0.0009**
$N_p = 25$	**86.3**	✗	41.4774	**0.0103**

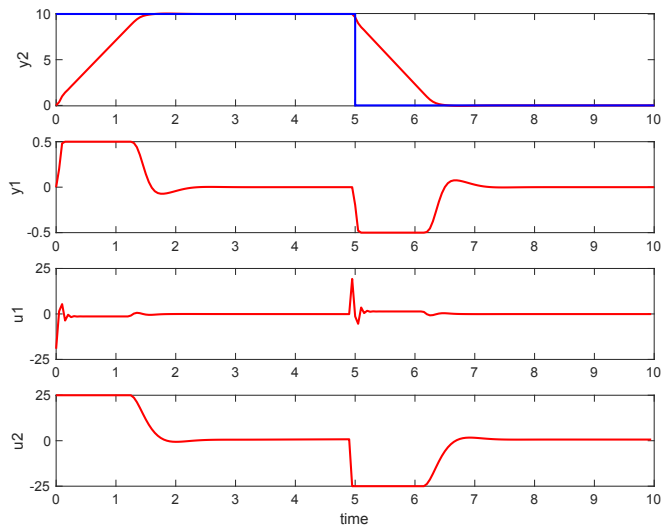

Fig. 4. Closed-loop performance of AFTI-16 under the MPC prediction horizon $N_p = 10$ setting and EIQP ($\epsilon = 10^{-8}$).

B. Real-time CLF-HOCBF-QP applications

This subsection uses EIQP to solve High-Order Control Barrier Function (HOCBF)-based QP [55] and to tell us what the minimum feedback time should be.

HOCBF has been shown to be capable of transforming nonlinear optimizations for nonlinear systems and constraints into QPs with formal guarantees [56]. We consider the Adaptive Cruise Control (ACC) problem with nonlinear vehicle dynamics [57]

$$\underbrace{\begin{bmatrix} \dot{z} \\ \dot{v} \end{bmatrix}}_{\dot{\boldsymbol{x}}} = \underbrace{\begin{bmatrix} v_p - v \\ -\frac{1}{M}F_r(v) \end{bmatrix}}_{f(\boldsymbol{x})} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{M} \end{bmatrix}}_{g(\boldsymbol{x})} u \tag{35}$$

where z denotes the distance between the ego vehicle and its preceding vehicle; v, v_p denote the speed of the ego and its preceding vehicle, respectively; u is the control (acceleration) of the ego vehicle; M is the mass of the ego vehicle, and $F_r(v) = f_0 \mathrm{sgn}(v) + f_1 v + f_2 v^2$, where $f_0 > 0$, $f_1 > 0$, and $f_2 > 0$ are scalars. The first term in $F_r(v(t))$ denotes the coulomb friction force, the second term denotes the viscous friction force, and the last term denotes the aerodynamic drag.

Constraint 1 (Control bounds): There are constraints on control for the ego vehicle, i.e.,

$$-c_d M g \leq u \leq c_a M g, \tag{36}$$

where $c_d > 0$ and $c_a > 0$ are deceleration and acceleration coefficients, respectively, and g is the gravity constant.

Constraint 2 (Safety constraint): The distance z is required to satisfy

$$z \geq \delta, \tag{37}$$

where $\delta > 0$ is determined by the length of the two vehicles.

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TABLE II
SIMULATION PARAMETERS FOR ACC

Parameter	Value	Units
δ	10	m
v_d	24	m/s
M	1650	kg
g	9.81	m/s^2
f_0	0.1	N
f_1	5	Ns/m
f_2	0.25	Ns^2/m
ϵ_{acc}	10	unitless
c_a, c_d	0.4	unitless
p_{acc}	1	unitless

Objective 1 (Desired speed): The ego vehicle always attempts to achieve a desired speed $v_d > 0$.

Objective 2 (Minimum acceleration): The control input effort of the ego vehicle,

$$J_i(u_i(t)) = \int_0^T \left(\frac{u(t) - F_r(v(t))}{M} \right)^2 dt, \tag{38}$$

should be minimized, where $T > 0$ is the final time.

Problem: The ACC problem is to determine control laws to achieve Objectives 1 and 2 subject to Constraints 1 and 2 for the ego vehicle governed by dynamics (35).

We use the Control Lyapunov Function (CLF) [58] to achieve the desired speed and use the HOCBF to enforce the safety constraint (37). Then the ACC problem can be transformed into the QP that is solved at every time step:

$$\boldsymbol{u}^*(t) = \arg\min_{\boldsymbol{u}(t)} \frac{1}{2} \boldsymbol{u}(t)^\top H \boldsymbol{u}(t) + F^\top \boldsymbol{u}(t) \tag{39}$$

$$\boldsymbol{u}(t) = \begin{bmatrix} u(t) \\ \delta_{acc}(t) \end{bmatrix}, H = \begin{bmatrix} \frac{2}{M^2} & 0 \\ 0 & 2p_{acc} \end{bmatrix}, F = \begin{bmatrix} \frac{-2F_r(v(t))}{M^2} \\ 0 \end{bmatrix}.$$

subject to

$$\begin{aligned} &L_f^2 b(\boldsymbol{x}) + L_g L_f b(\boldsymbol{x}) u + 2L_f b(\boldsymbol{x}) + b(\boldsymbol{x}) \geq 0, \\ &L_f V(\boldsymbol{x}) + L_g V(\boldsymbol{x}) u + \epsilon_{acc} V(\boldsymbol{x}) \leq \delta_{acc}, \\ &-c_d M g \leq u \leq c_a M g, \end{aligned}$$

where $b(\boldsymbol{x}) = z - \delta$ is the HOCBF, $V(\boldsymbol{x}) = (v - v_d)^2$ is the CLF; L_f, L_g denotes the Lie derivative along f, g, respectively; $\epsilon_{acc} > 0$; and δ_{acc} is a relaxation variable that ensures the CLF will not conflict with the HOCBF constraint.

The simulation parameters are listed in Table II. The initial conditions are $z(0) = 100\,\mathrm{m}$, $v(0) = 20\,\mathrm{m/s}$, and $v_p(t) = 13.89\,\mathrm{m/s},\ \forall t \in [0, T]$.

Simulation results. As the Hessian matrix of this CLF-HOCBF-QP (39) has a condition number of 2.7225×10^6. As demonstrated in Subsection V-C, our current EIQP implementation is not sufficiently robust for achieving high accuracy on ill-conditioned QPs. In this CLF-HOCBF-QP (39) example, our EIQP has to choose $\epsilon = 10^{-14}$ to achieve zero-violations in closed-loop. EIQP (with $\epsilon = 10^{-14}$) is compared against "Quadprog", "OSQP", "SCSv3", and "Clarabel" (all under default settings). As shown in Tab. III, "SCSv3" solver violates the safety guarantee, since its $b(\mathbf{x}(t)) \geq 0$ does not hold. Fig. 5 also shows that "SCSv3" and "Clarabel" solvers produce non-smooth input profiles, whereas our proposed "EIQP", "OSQP", and "Quadprog" do not and have the same simulation results.

Tab. III also reports the execution time of different solvers. Since this CLF-HOCBF-QP (39) is very small-scale ($n_z = 2, n_b = 3$), the observed differences in computation time are likely dominated by interface overheads of the general-purpose solvers. Nevertheless, our proposed EIQP has the **smallest** standard deviation of execution time, which is consistent with our theoretical result: EIQP performs *exact* and *fixed* iterations. This certifies that EIQP's execution time is below 0.075 [ms], enabling our CLF-HOCBF-QP controller to operate at a maximum frequency of 13.3 kHz. This property is particularly useful for determining the minimum discretization (feedback) time when designing continuous CLF-CBF-QP controllers, as discussed in [59].

TABLE III
COMPARISON BETWEEN DIFFERENT SOLVERS IN CLF-HOCBF-QP

Solver	Execution time [ms]	HOCBF $b(\boldsymbol{x}(T))$
Quadprog	0.318±0.412	1.825×10^{-7}
OSQP	0.361±0.196	1.825×10^{-7}
SCSv3	0.286±1.807	$\mathbf{-6.3981 \times 10^{-7}}$
Clarabel	0.0261±0.0284	1.9005×10^{-7}
EIQP	0.0624± **0.003**	4.4704×10^{-8}

VII. CONCLUSION

This article proposes an execution-time-certified and infeasibility-detecting solver for convex QP (including LP), which is based on an infeasible IPM algorithm. The homogeneous formulation of convex QP's KKT condition is adopted to achieve the infeasibility-detecting capability. Moreover, by exploiting this homogeneous formulation, our proposed infeasible IPM algorithm can achieve the best theoretical $O(\sqrt{n})$ iteration complexity that feasible IPM algorithms enjoy. The iteration complexity is proved to be *exact* (rather than an upper bound), *simple to calculate*, and *data independent*, with the value $\left\lceil \frac{\log(\frac{n+1}{\epsilon})}{-\log(1-\frac{0.414213}{\sqrt{n+1}})} \right\rceil$, making it appealing to certify the execution time of online time-varying convex QPs and LPs.

The homogeneous formulation comes at the cost of ill-conditioned linear systems to be solved; we achieved numerical robustness in our current implementation by employing LU decomposition. Future work includes: *i)* developing a good preconditioner for solving the linear system at each iteration, to enable the faster decomposition to be used; *ii)* taking advantage of the full Newton step feature to parallelize the predetermined sequential iterations, which can greatly improve the computational efficiency.

VIII. ACKNOWLEDGEMENT

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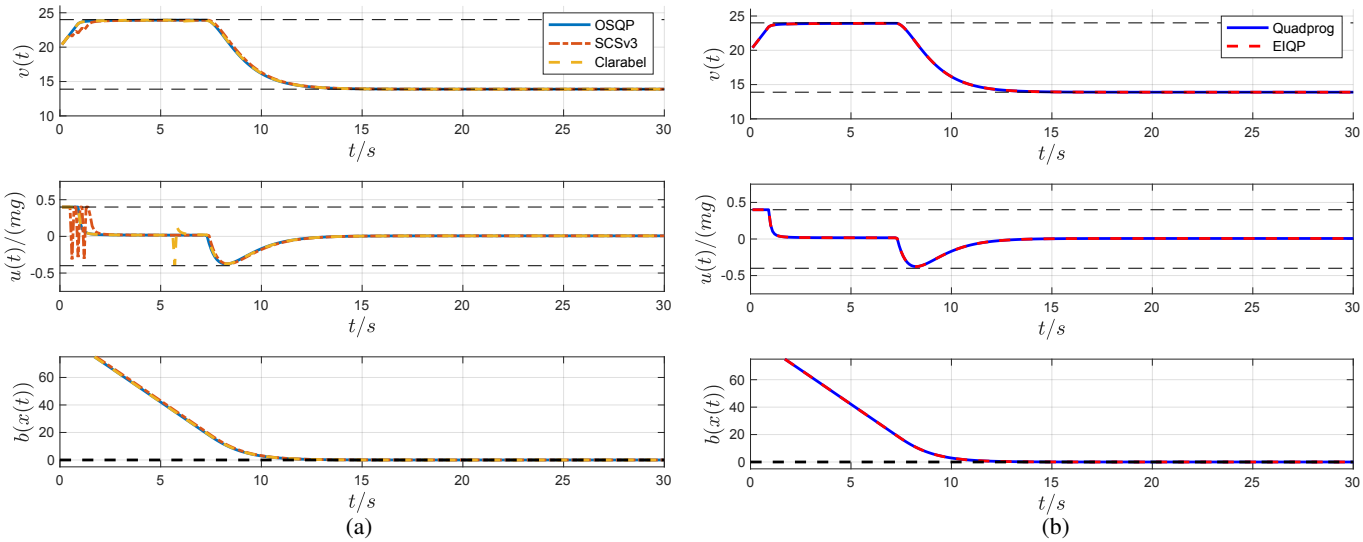

Fig. 5. State profiles and HOCBFs for different solvers. $b(\boldsymbol{x}(t)) \geq 0$ denotes safety guarantees. (a) OSQP, SCSv3, and Clarabel; (b) Quadprog and EIQP.

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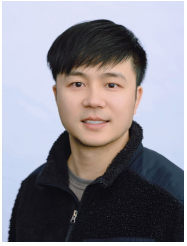

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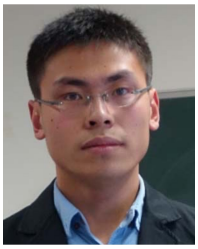

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